

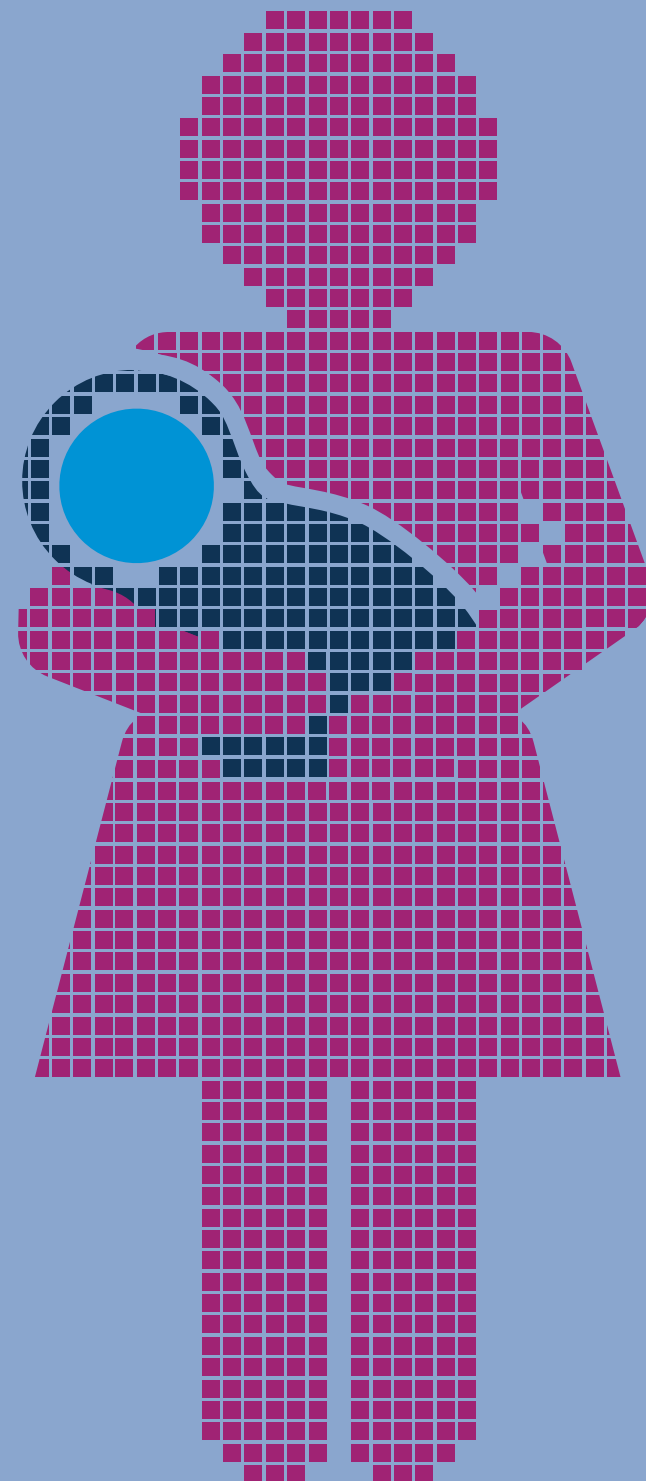
Digital adaptation kit for postnatal care

Operational requirements for implementing
WHO recommendations in digital systems



World Health
Organization

human
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UNDP • UNFPA • UNICEF • WHO • WORLD BANK



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Digital adaptation kit for postnatal care: operational requirements for implementing WHO recommendations in digital systems
(SMART Guidelines collection)

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Indicators and performance metrics	https://smart.who.int/dak-pnc/indicators.html
Functional and non-functional requirements	https://smart.who.int/dak-pnc/system-requirements.html

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Abbreviations

ANC	antenatal care	PCPOSS	person-centred point of service system(s)
CHW	community health worker	PNC	postnatal care
DAK	digital adaptation kit	SIDS	sudden infant death syndrome
HMIS	health management information system	SMART	standards-based, machine-readable, adaptive, requirements-based and testable
ICD	International Classification of Diseases	SNOMED	Systematized Nomenclature of Medicine
ID	identification	SRH	Sexual and Reproductive Health and Research (WHO department)
IUD	intrauterine device	TB	tuberculosis
MCA	Maternal, Newborn, Child and Adolescent Health and Ageing (WHO department)	UHC	universal health coverage
MOH	ministry of health	WHO	World Health Organization

Glossary

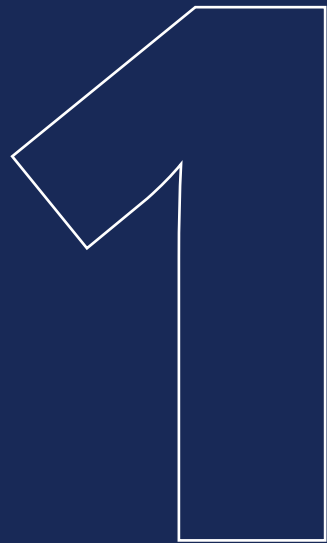
Note: Terms in definitions also defined in this glossary shown in *italics*.

Business process	A set of related activities or <i>tasks</i> performed together to achieve the objectives of the health programme area, such as registration, counselling and referrals (1,2).
Data dictionary	A centralized repository of information about the <i>data elements</i> that contains their definition, relationships, origin, usage and type of data. For this digital adaptation kit, the data dictionary is provided as a spreadsheet.
Data element	A unit of data that has specific and precise meaning.
Decision-support logic	A set of decision rules for standard and exceptional cases that is separate from the <i>business process</i> . This will help to reduce the complexity of the <i>business process</i> depiction without losing the detail necessary for coding the rules required for system functionality.
Decision support (for health workers)	Digitized job aids that combine an individual's health information with the health worker's knowledge and clinical protocols to assist health workers in making diagnosis and treatment decisions (3,4).
Decision-support table	Semi-structured way to depict each discrete decision that will need to be embedded in the system. Depending on the complexity of the clinical guidelines, there will likely be multiple decision-support tables.
Digital health	The systematic application of information and communications technologies, computer science and data to support informed decision-making by individuals, the health workforce and health systems, to strengthen resilience to disease and improve health and wellness (1).
Digital tracking	The use of a digitized record to capture and store clients' health information to enable follow-up of their health status and services received, also referred to as longitudinal tracking of person's health status and services. This may include digital forms of paper-based registers and case management logs within specific target populations, as well as electronic medical records linked to uniquely identified individuals (3,4).
Functional requirement	Capabilities the system must have in order to meet the end-users' needs and achieve <i>tasks</i> within the <i>business process</i> .
Health information system	A system that integrates data collection, processing, reporting and use of the information necessary for improving health service effectiveness and efficiency through better management at all levels of health services (5).
Health management information system (HMIS)	An information system specifically designed to assist in the management and planning of health programmes, as opposed to delivery of care (5).
Interoperability	The ability of different applications to access, exchange, integrate and use data in a coordinated manner through the use of shared application interfaces and standards, within and across organizational, regional and national boundaries, to provide timely and seamless portability of information and optimize health outcomes.
Non-functional requirement	General attribute or feature of the digital system to ensure usability and overcome technical and physical constraints. Examples of non-functional requirements include ability to work offline, multiple language settings and password protection.
Persona	A generic aggregate description of a person involved in or benefitting from a health programme.
Person-centred point of service systems (PCPOSS)	A digital system that facilitates the provision and delivery of health services to individuals (i.e. persons, clients, patients, health-service users) at the point of service or point of care. This includes software capabilities that enable health workers to access, record and update individuals' health information as well as software capabilities that enable health workers to communicate with individuals. PCPOSS encompass various services and application types, including community-based information systems, decision-support systems, electronic medical record system and personal health records (4).

Standard	In software, a standard is a specification used in digital application development that has been established, approved and published by an authoritative organization. These rules allow information to be shared and processed in a uniform, consistent manner independent of a particular application.
Task	A specific action in a <i>business process</i> .
Terminologies	For clinical care, terminologies are structured vocabularies covering health-related concepts – such as diseases, diagnoses, laboratory tests and treatments – to enable the storage, analysis and exchange of data in a consistent and standard way (6).
Workflow	A visual representation of the progression of activities (<i>tasks</i> , events, decision points) in a logical flow illustrating the interactions within the <i>business process</i> (2).

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Overview of digital adaptation kits

Background

Digital health – defined broadly as the systematic application of information and communications technologies, computer science and data to support informed decision-making by individuals, the health workforce and health systems, to strengthen resilience to disease and improve health and wellness (1) – is increasingly being applied as an essential enabler for health-service delivery and accountability. Ministries of health have recognized the value of digital health as articulated within a World Health Assembly resolution (2) and the *Global strategy on digital health* (3). Likewise, donors have advocated for the rational use of digital tools as part of efforts to expand coverage and quality of services, as well as promote data use and monitoring efforts (4–6). Despite the investments into and abundance of digital systems, there is often limited transparency in the health data and logic contained in these digital tools, or relationship with evidence-based clinical or public health recommendations, which not only undermines the credibility of such systems, but also impedes opportunities for interoperability and threatens potential for continuity of care, particularly with regards to the informational component (7–10).

Evidence-based recommendations, such as those featured in World Health Organization (WHO) guidelines, establish standards of care and offer a reference point for informing the content of digital systems that countries adopt. However, guidelines are often only available in a narrative format that requires a resource-intensive process to be elaborated into the specifications needed for digital systems. This translation of guidelines for digital systems can result in subjective interpretation for implementers and software vendors, which can lead to inconsistencies or inability to verify the content within these systems, potentially leading to adverse health outcomes and other unintended effects (8,11–13). Where digital systems exist, the documentation of the underlying data and content may be unavailable or proprietary, requiring governments to start from scratch and expend additional resources each time they intend to deploy such a system. This lack of documentation of the health content can lead to dependence on one vendor and haphazard deployments that are unscalable or difficult to replicate across different settings.

To ensure countries can effectively benefit from digital health investments, the Standards-based, Machine-readable, Adaptive, Requirements-based and Testable (SMART) guidelines are designed to facilitate the accurate reflection of WHO’s clinical, public health and data use guidelines within the digital systems countries are adopting (14). “Digital adaptation kits” (DAKs) are the operational layer of the SMART guidelines framework and they provide software-neutral, standardized documentation that distils clinical, public health and data use guidance into a format that can be transparently incorporated into digital systems (9). Although digital implementations comprise multiple factors – including the (i) health domain data and content, (ii) digital intervention or functionality, and (iii) digital application or communication channel for delivering the digital intervention – SMART guidelines, and therefore DAKs, focus primarily on ensuring the validity of the health content (see Fig. 1) (1,15).

Accordingly, DAKs provide the generic content requirements that should be housed within digital systems in a human-readable format, independently of a specific software application and with the intention that countries can customize them to local needs.

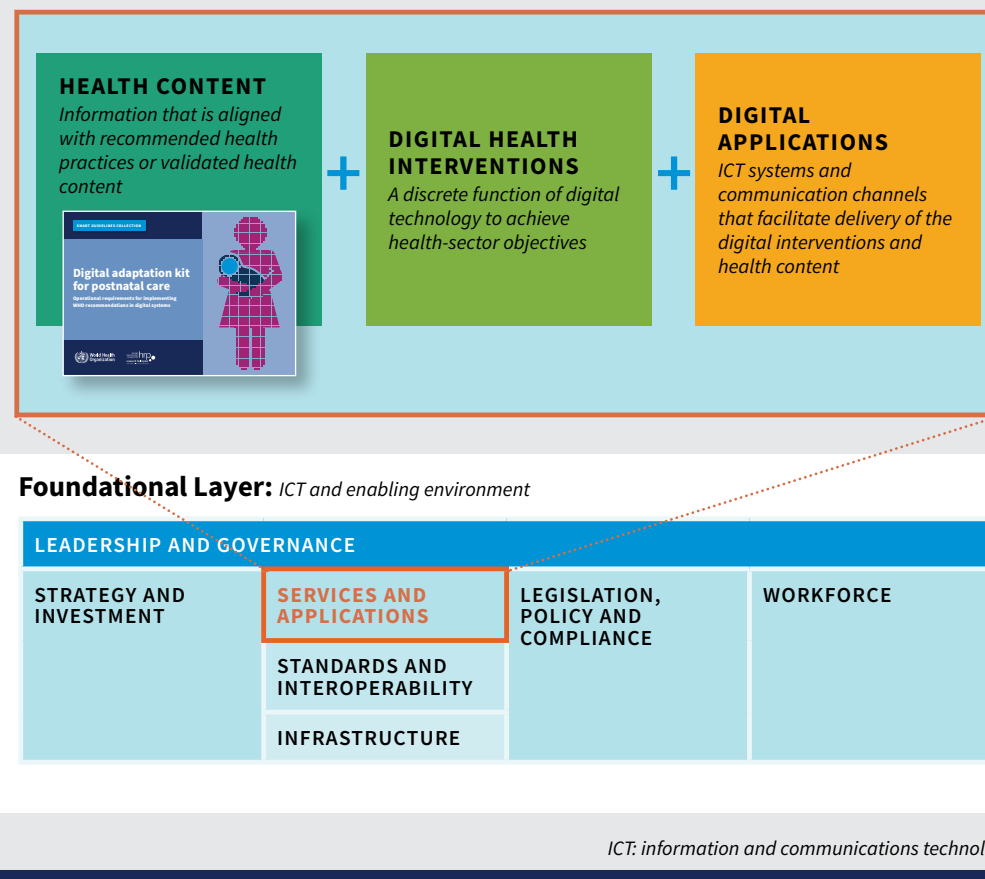
For this particular DAK, the requirements are based on systems that provide the functionalities of person-centred point of service systems (PCPOSS) (Box 1), and include components such as personas, workflows, core data elements, decision-support algorithms and scheduling logic. Operational outputs, such as

spreadsheets of the data dictionary and the detailed decision-support algorithms, are included as part of the DAK as practical resources that implementers can use as starting points when developing digital systems. Furthermore, data components within the DAK are mapped to standards-based terminology, such as the International Classification of Diseases (ICD), International Classification of Functioning, Disability and Health (ICF), International Classification of Health Interventions (ICHI), Logical Observation Identifiers Names and Codes (LOINC) and Systematized Nomenclature of Medicine (SNOMED), to facilitate interoperability. **Box 2** provides context-specific recommendation for the use of an integrated PCPOSS that provides both digital tracking of client's health status and services and decision support.

The DAKs follow a modular approach in detailing the data and content requirements for a specific health programme area – such as antenatal care (ANC) (16), family planning (17) and HIV (18) – among the different health areas for which DAKs have been developed. **This DAK focuses on providing the content requirements for PCPOSS used in any health-care setting by health workers providing postnatal care (PNC) and is based on the WHO PNC guideline: WHO recommendations on maternal and newborn care for a positive postnatal experience** (19).

Fig. 1

Digital adaptation kits and their role in digital health implementations



Source: Adapted from WHO (15).

Box 1

What are person-centred point of service systems?

A person-centred point of service system (PCPOSS) is digital in nature and facilitates the provision and delivery of health services to individuals (i.e. persons, clients, patients and health-service users) at the point of care. A PCPOSS includes software capabilities that enable health workers to access, record and update individuals' health information, and interactively communicate with them. PCPOSS encompass various services and application types including:

- » **community-based information systems** that “facilitate data collection and use at the community level. These applications are utilized by community-based workers who provide health promotion and disease prevention activities”;
- » **decision-support systems** – digital “tools which combine medical information databases and algorithms with patient specific data. They are intended to provide healthcare professionals and/or users with recommendations for diagnosis, prognosis, monitoring and treatment of individual patients”;
- » **electronic health record system** – a “secure, online system that holds information about people’s health and clinical care and is managed by health workers”;
- » **personal health record** – the “record of an individual’s health information in a structured digital format for a set of defined use cases over which the person has agency”.

PCPOSS end-users may include all health worker occupational groups operating at all care levels, including those operating outside of formal health facilities (e.g. community health workers and health volunteers).

Source: WHO (20).

Based on the principle of “collect once, use for many purposes”, this DAK outlines the requirements of a PCPOSS that facilitates the use of more reliable source data for aggregate indicators at management levels by providing access to primary data collected directly at the point of care, allowing a shift away from the need for aggregate indicators to be reported separately and for investments in paper-based or digital indicator reporting systems. Data collected for the purposes of service delivery can also be used to calculate aggregate indicators required for reporting and accountability, including monitoring provider, stock and system performance. Factors affecting overall health-system performance can thus be highlighted more promptly and accurately while reducing the clerical burden on health workers.

Digital health interventions incorporated into this digital adaptation kit

In the context of this DAK, the PCPOSS focuses on the following digital health interventions (20):

- » enrolling person(s) for health services/clinical care plan
- » longitudinal tracking of health-service user's health status and services
- » managing person-centred structured clinical records
- » managing person-centred unstructured clinical records (e.g. notes, images, documents)
- » providing prompts and alerts according to protocols
- » providing checklists according to protocols
- » screening persons by risk or other health status
- » transmitting targeted alerts and reminders to health-service user(s)
- » identifying persons in need of services
- » communication and performance feedback to health workers
- » routine health indicator data collection and management
- » data storage and aggregation
- » data synthesis and visualizations.

Box 2

WHO recommendation related to PCPOSS

WHO has provided the following context-specific recommendation for the use of an integrated system that provides both digital tracking of client's health status and services and decision support.

Source: WHO (15).

Effective coverage
Accountability
coverage

Digital tracking of clients' health status and services (digital tracking) combined with decision support

RECOMMENDATION 8: WHO recommends digital tracking of clients' health status and services, combined with decision support under these conditions:

- ▶ in settings where the health system can support the implementation of these intervention components in an integrated manner; and
- ▶ for tasks that are already defined as within the scope of practice for the health worker.

(Recommended only in specific contexts or conditions)

Digital adaptation kits within a strategic vision for SMART guidelines

The operational and standardized documentation reflected within the DAKs represent one of the steps within a broader vision of **S**tandards-based, **M**achine-readable, **A**daptive, **R**equirements-based and **T**estable (SMART) guidelines (14). SMART guidelines aim to maximize health impact through improved fidelity and uptake of recommendations within standards-based digital systems through a systematic process for transforming guideline development, delivery and application (13,21). Within this vision, DAKs serve as a prerequisite for developing computable, or machine-readable, guidelines, as well as executable reference software and advanced analytics for precision health. **Figure 2** provides an overview of the different layers of the SMART guidelines continuum and where DAKs fit within this strategy (21).

Objectives

This DAK focuses on PNC and aims to provide a common language across various audiences – maternal, newborn/infant and child health and other programme managers, software developers and implementers of digital systems – to ensure a common understanding of the appropriate health information content within a defined health programme area, as a mechanism to catalyse the effective use of these digital systems. The key objectives of the DAK are:

- » to ensure adherence to WHO clinical, public health and data use guidelines, and facilitate consistency of the health content that is used to inform the development of a PCPOSS;
- » to enable both health programme leads and digital health teams (including software developers) to have a joint understanding of the health content within the digital system, with a transparent mechanism to review the validity and accuracy of the health content; and
- » to provide a starting point of the core data elements and decision-support logic that should be included within PCPOSS for PNC.

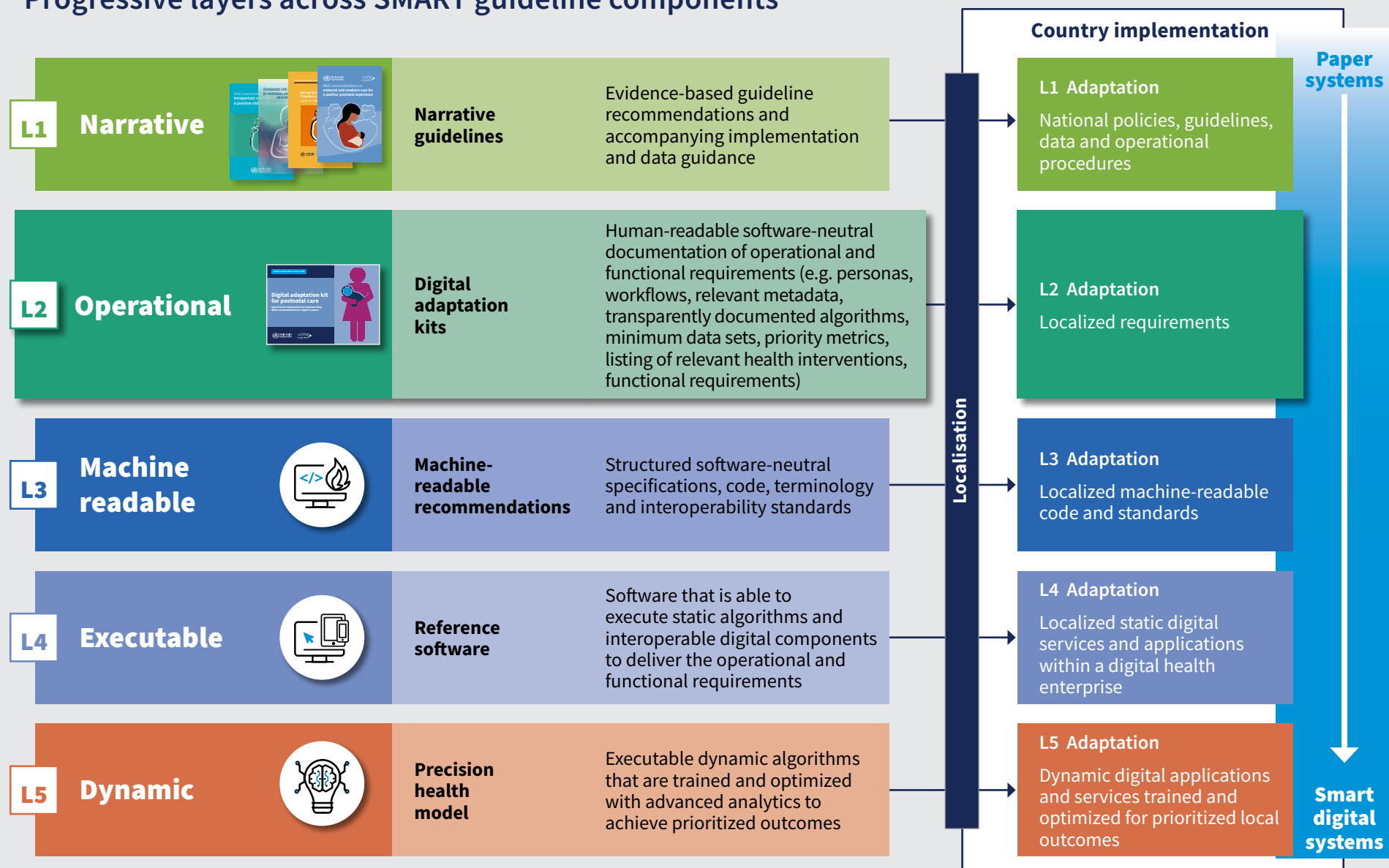
Information detailed in this DAK reflects generic workflow processes, data and decision-support algorithms, as derived from the *WHO recommendations on maternal and newborn care for a positive postnatal experience* (19) and other related WHO documents described below (see [section 1.2](#)). **Note that the outputs of the DAKs are intentionally generic and will need to be contextualized to local policies and requirements.** In fact, the WHO PNC recommendations need to be adapted, resourced and integrated into maternal, newborn/infant and child health programmes that reflect the needs of different countries, local contexts, and individual women, newborns/infants, parents, caregivers and families (19).

For this reason, this DAK has been developed under certain assumptions.

- » The contents of the WHO PNC recommendations that have been translated into this DAK are intended for women or adolescent girls after childbirth, and their newborns/infants, who have no apparent risk factors or illness, and who otherwise appear to be healthy.
- » The contents of the WHO PNC recommendations that have been translated into this DAK may not cover the additional health and social needs of some women and their newborns/infants; for example, additional management of women and newborns/infants who develop complications in the postnatal period, or of those with high-risk pregnancies or who presented complications during labour and childbirth or who require specialized care.

Fig. 2

Progressive layers across SMART guideline components



Source: adapted from Mehl et al. (21).

- » The essential postnatal care package included in the WHO PNC recommendations supporting a minimum of four contacts following childbirth and extending up to six weeks (42 days) is reflected in this DAK. The first contact refers to continued care in the health facility for at least the first 24 hours after birth or a first postnatal contact within the first 24 hours for a home birth, while the three additional PNC contacts occur between 48 and 72 hours, between 7 and 14 days, and during week six after birth; it is recommended that the contact that occurs between 48 and 72 hours be conducted in the home, to allow the health worker to see the mother and newborn in their home environment.
- » The contents of the WHO PNC recommendations translated into this DAK are relevant to any health facility or community-based setting, regardless of the place of birth and whether the health system is set up for a home visit in the first week or if the woman and newborn need to seek routine outpatient postnatal care at the health facility or in the community.

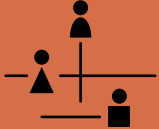
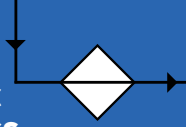
The contents of the WHO PNC recommendations translated into this DAK are complementary to existing WHO guidance on ANC, on the management of complications during pregnancy and childbirth, and on the care of the mother and newborn immediately after birth. Considering antenatal, intrapartum and postnatal care as a continuum is key to ensuring the best outcomes for maternal and newborn/infant health. For this reason, the WHO PNC recommendations may also refer to interventions in the antenatal period, encompass elements of immediate care or include assessments of woman and newborn/infant to detect complications although these topics are addressed in other WHO guidance. The overarching aim of each PNC contact is to provide women, newborns/infants, parents and caregivers with respectful, individualized, person-centred care regardless of the resource setting and of the specific set-up of the health system (19).

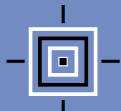
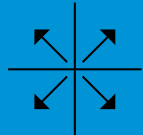
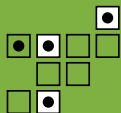
DAKs have also been developed for ANC (16), family planning (17), HIV (18), tuberculosis (TB) (22), child health in humanitarian emergencies (23) and immunization (24). This approach is being expanded to additional health domains, such as self-monitoring of blood pressure during pregnancy, birth defects surveillance and intrapartum care. All of these DAKs work towards a comprehensive approach for standardized software requirements for primary health care settings.

Components of a digital adaptation kit

The DAK comprises eight interlinked components: (1) health interventions and associated recommendations; (2) generic personas; (3) user scenarios; (4) generic business processes and workflows; (5) core data elements; (6) decision-support logic; (7) indicators and reporting requirements; and (8) high-level functional and non-functional requirements. **Table 1** provides an overview of each of the components of the DAK, which this document elaborates. All information within the adaptation kit represents a generic starting point, which should then be adapted according to the needs of different contexts. **Box 2** provides notation guidance that is used in the implementation tools.

Table 1. Components of a digital adaptation kit

Component	Description	Purpose	Outputs	Adaptation needed
1. Health interventions and recommendations 	Overview of the health interventions and WHO recommendations included within this DAK. DAKs are meant to be a repackaging and integration of WHO guidelines and guidance documents in a particular health domain. The list of health interventions is drawn from the universal health coverage (UHC) menu of interventions compiled by WHO (25).	Setting the stage – To understand how this DAK would be applied to a digital tracking and person-centred point of service systems in the context of specific health programmes and interventions	» List of related health interventions based on the WHO UHC essential interventions (25) » List of related WHO recommendations based on guidelines and guidance documents	» Contextualization to reflect current or planned national policies
2. Generic personas 	Depiction of the end-users, supervisors and related stakeholders who would be interacting with the digital system or involved in the care pathway.	Contextualization – To understand the wants, needs and constraints of the end-users	» Description, competencies and essential interventions performed by targeted health worker personas	» Greater specification and details of the end-users based on real people (e.g. health workers) in a given context » High-level information to describe the provider of the health service (e.g. the general background, roles and responsibilities, motivations, challenges and environmental factors)
3. User scenarios 	Narratives that describe how the different personas may interact with each other. The user scenarios are only illustrative and are intended to give an idea of a typical workflow.	Contextualization – To understand how the system would be used, and how it would fit into existing workflows	» Example narrative of how the targeted personas may interact with each other during a workflow	» Greater specification and details on the real needs of end-users in a given context
4. Generic business processes and workflows 	A business process is a set of related activities or tasks performed together to achieve the objectives of the health programme area, such as registration, counselling, referral (1,26). Workflows are a visual representation of the progression of activities (tasks, decision points, interactions) that are performed within the business process (1,26).	Contextualization and system design – To understand how the digital system would fit into existing workflows and how best to design the system for that purpose	» Overview matrix presenting the key processes in PNC » Workflows for identified business processes with annotations	» Customization of the workflows, which can include additional forks, alternative pathways or entirely new workflows

Component	Description	Purpose	Outputs	Adaptation needed
5. Core data elements 	Data elements required throughout the different points of the workflow. These data elements are mapped to the International Classification of Diseases version 11 (ICD-11) codes and other established concept mapping standards to ensure the data dictionary is compatible with other digital systems.	System design and interoperability – To know which data elements need to be logged and how they map to other standard terminologies (e.g. ICD, Systematized Nomenclature of Medicine [SNOMED]) for interoperability with other standards-based systems	» List of data elements » Link to data dictionary with detailed data specifications in spreadsheet format (available here)	» Translation of “data labels” into the local language and additional data elements created depending on the context
6. Decision-support logic 	Decision-support logic and algorithms to support appropriate service delivery in accordance with WHO clinical, public health and data use guidelines.	System design and adherence to recommended clinical practice – To know what underlying logic needs to be coded into the system	» List of decisions that need to be made throughout the encounter » Link to decision-support tables in a spreadsheet format with inputs, outputs and triggers for each decision-support logic (available here) » Scheduling logic for services (available here)	» Change of specific thresholds or triggers in a logic (IF/THEN) statement, e.g. body mass index (BMI) cut-off, age trigger for “youth friendly” services » Additional decision-support logic formulas depending on the context
7. Indicators and performance metrics 	Core set of indicators that need to be aggregated for decision-making, performance metrics based on information captured in this system. These indicators and metrics are based on data that can feasibly be captured from a routine digital system, rather than survey-based tools.	System design and adherence to recommended health monitoring practices – To know what calculations and secondary data use are needed for the system, based on the principle of “collect once, use many” (27)	» Indicators table with numerator and denominator of data elements for calculation, along with appropriate disaggregation (available here)	» Changing calculation formulas of indicators » Adding indicators » Changing the definition of the primary data elements used to calculate the indicator based on data available
8. Functional and non-functional requirements 	List of core functions and capabilities the system must have to meet the end-users’ needs and achieve tasks within the business processes.	System design – To know what the system should be able to do	» Table of functional and non-functional requirements with the intended end-user of each requirement, as well as why that user needs that functionality in the system (available here)	» Adding or reducing functions and system capabilities based on budget and end-user needs and preferences

Throughout the DAK, there are identification (ID) numbers to simplify tracking and referencing of each of the components. Note that the DAK represents an overview across the different components, while the comprehensive and complete outputs of each component (e.g. data dictionary, decision-support tables) are included in appended spreadsheets. The notation guidance is as follows.

Box 3

Notation guidance

Component 1: Health interventions and recommendations

No notations used

Component 2: Generic personas

No notations used

Component 3: User scenarios

No notations used

Component 4: Business processes and workflows

- » Each workflow should have a “Process name” and a corresponding letter
- » Each workflow should also have a “**Process ID**” that should be structured “**Abbreviated health domain**” (e.g. PNC).“**Corresponding letter for the process**” (e.g. A), e.g. PNC.A
- » Each activity in the workflow should be numbered with an “**Activity ID**” that should be structured “**Process ID**” from above “**Activity number**”, e.g. PNC.B7

Component 5: Core data elements (data dictionary)

- » Each data element should have a running number and a “**Data element (DE) ID**” that should be structured “**Activity ID**” (e.g. PNC.B7).“DE”.“Sequential number of the data element” (e.g. PNC.B7.DE.1, PNC.B7.DE.2)

Component 6: Decision-support logic

- » Each decision-support logic table should have a running number and a “**Decision-support table (DT) ID**” that should be structured “**Activity ID**” (e.g. PNC.D2).“DT”.“Sequential number of the decision-support table” (e.g. PNC.DT.1, PNC.DT.2)

Component 7: Indicators and performance metrics

- » Each indicator should have an “**Indicator ID**” that should be structured “**Abbreviated health domain**” (e.g. PNC).“IND”.“Sequential number of the indicator” (e.g. PNC.IND.1, PNC.IND.2)

Component 8: High-level system requirements

- » Each functional requirement should have a “**Functional requirement ID**” that should be structured “**Abbreviated health domain**” (e.g. PNC).“REQ”.“Sequential number of the functional requirement” (e.g. PNC.REQ.1, PNC.REQ.1)
- » Each non-functional requirement should have a “**Non-functional requirement ID**” that should be structured “**Abbreviated health domain**” (e.g. PNC).“NFXNREQ”.“Sequential number of non-functional requirement” (e.g. PNC.NFXNREQ.1, PNC.NFXNREQ.2)

How this digital adaptation kit was developed

A mapping of existing WHO guidelines, guidance and tools considered relevant for the development of a DAK dedicated to PNC was carried out. Key resources that were identified included the PNC guidelines and additional relevant guidance documents and associated handbooks (see [section 1.2](#)). A desk review of the above resources was conducted, where the recommendations from clinical guidelines were extracted and synthesized to form the components of the DAK for PNC. This process also guided the development of the workflows for key PNC programmatic processes, decision-support logic algorithms and a data dictionary. The indicators included in the PNC DAK were informed by *Analysis and use of health facility data: guidance for maternal, newborn, child and adolescent health programme managers* (28). The data elements required to calculate the indicators and to build the algorithms in the decision-support logic were based on the relevant clinical guidelines and guidance documents. The data elements were mapped onto standardized terminologies and classifications by a medical terminologist to facilitate the adoption of interoperability standards into digital systems. Components of the PNC DAK were further refined through a series of consultations with external experts and stakeholders and relevant teams from the WHO Departments of Maternal, Newborn, Child and Adolescent Health and Ageing (MCA) and Sexual and Reproductive Health and Research (SRH), which were responsible for developing the WHO guidelines and guidance used to guide the development of this DAK; DAKs are derivative guidance documents that support the implementation of WHO guidelines and guidance in digital systems. Other published DAKs, including the teams responsible for their development, were consulted as needed to ensure alignment across the SMART Guidelines programme (14).

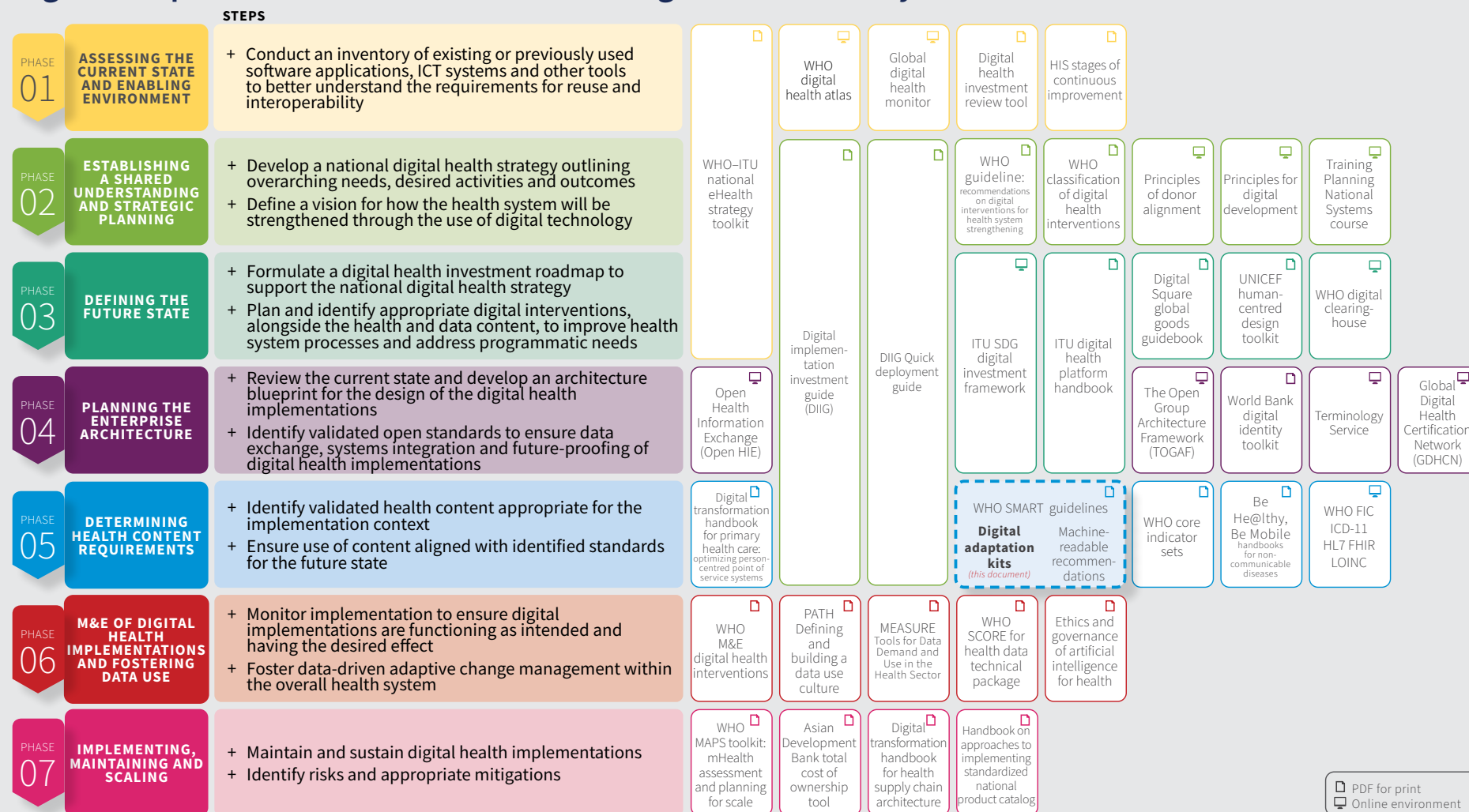
Individual, external contributors involved in the development and review of this DAK declared any potential conflicts of interest prior to providing their contributions. These declarations were reviewed and none of the contributors were found to have conflicting interests that would influence the content of this publication.

How to use this digital adaptation kit

Once the strategic vision for a point of service system has been developed, SMART guidelines can assist in determining the necessary health content requirements based on available guidance. In contexts where an overarching vision may not exist, users should first consult the WHO and International Telecommunication Union National eHealth strategy toolkit (29), WHO guidelines on digital interventions for health system strengthening (15), and the WHO Digital Implementation Investment Guide (DIIG) (1) to establish a better understanding of how to select and apply appropriate digital health interventions. [Figure 3](#) situates DAKs within the broader set of resources for planning and implementing digital health systems.

Fig. 3

Digital adaptation kits within the broader digital health ecosystem



FIC: Family of International Classifications; HIS: health information system; HL7: Health Level 7; ICD: International Classification of Diseases; ICT: information and communications technology; ITU: International Telecommunication Union; LOINC: Logical Observation Identifiers Names and Codes; M&E: monitoring and evaluation; MAPS: mHealth Assessment and Planning for Scale; SDG: Sustainable Development Goal; WHO: World Health Organization.

Source: adapted from WHO (1).

Target audience

The primary target audience for this DAK is health programme managers within the ministry of health (MOH), who will be working with their digital or health information systems counterparts in determining the health content requirements for PNC PCPOSS. The health programme manager is responsible for overseeing and monitoring the implementation of the clinical practices and policies for the health programme area, in this case PNC.

The DAK also equips individuals responsible for translating health-system processes and guidance documents for use within digital systems with the necessary components to kick-start the process of developing PCPOSS in a standards-compliant manner. These individuals are also known as business analysts who interface between health content experts and software development teams. Specifically, the adaptation kit contains key outputs, such as the data dictionary and decision-support algorithms, to ensure the validity and consistency of the health content with the PCPOSS.

Additionally, using this DAK requires a collaboration between health programme managers responsible for other relevant health domains and counterparts in digital health and health information systems. Although each DAK focuses on a particular health programme area (in this case PNC), the DAKs are envisioned to be used in a modular format and link to other health programme areas within primary health care settings, in an effort to support integration across services.

Scenarios for using the DAK

The DAK may be used across various scenarios, some of which are listed below.

Scenario 1: Incorporating WHO guideline content into existing digital systems

Countries that already have digital systems in place, such as electronic medical records and decision-support tools, may use the information in the DAK to cross-check whether the underlying content and data for specific health programme areas are aligned with WHO guidelines. Users of the DAK can identify and extract specific decision algorithms that would need to be incorporated into their existing digital systems. By reviewing this systematic documentation, health programme managers and implementers can more readily identify differences in workflows, data inputs and decision-support logics to examine the rationale for deviations and understanding local adaptations of guideline content.

Scenario 2: Transitioning from paper to person-centred point of service systems

Some countries may currently have paper-based systems that they would like to digitize. The process of optimizing paper-based client-level systems into digital records and decision support may be overwhelming. Users in this scenario may review the DAK as a starting point for streamlining the necessary data elements and decision support that should be in the optimized client-level digital system. Users may also then refer to the paper-based tools to determine whether there are missing fields or content that should also be included within the digital system.

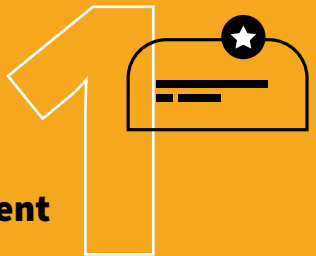
Users should also review the *WHO Digital transformation handbook for primary health care (30)*, which provides stepwise guidance on how to map data on paper-based forms into a digital system, including ways of accounting for data elements that are redundant or may not add value to the health system.

Scenario 3: Linking aggregate health management information system (HMIS, e.g. DHIS2) and person-centred point of service systems	In some instances, countries may already have a digital system for aggregate reporting (e.g. an HMIS) but may not yet have implemented digital systems that function at the point of service. The DAK can guide the development of a PCPOSS that includes an electronic medical record system, a community-based information system, and decision-support systems that operates at the point of service, or point of care, and ensures that there are linkages between the data collected at the service-delivery levels (e.g. community or facility level) and at the aggregate levels. As such, a component of the DAK provides aggregate indicators that can be derived from individual-level data to facilitate the linkage between these different levels. Complementary guidance dedicated specifically to aggregate-level data, such as <i>Analysis and use of health facility data: guidance for maternal, newborn, child and adolescent health programme managers (28)</i> should also be consulted for supporting the use of routine data at the facility management and district levels.
Scenario 4: Leveraging data standards to promote interoperability and integrated systems	This DAK includes data elements mapped to ICD codes, and other standards, to support the design of interoperable systems. The data dictionary provides the necessary codes for different data elements, thus reducing the time for implementers to incorporate these global standards into the design of their digital systems.
Scenario 5 Linking point-of-service systems used at the point of care by health workers to personal health records systems held by individuals	A critical part of service delivery in any health domain relies on engaging with clients. Digital interventions aimed at clients themselves – such as on-demand information services, targeted client communication (e.g. transmitting health information and reminders), reporting of health-system feedback by clients on the quality of care, accessing their own medical records/home-based records, and self-monitoring of their health and diagnostic data (16) – are all emerging approaches for complementing the services provided by health workers. Specifically for immunizations, clients will need to have access to their immunization history for purposes of continuity of care and/or providing proof of vaccination for purposes not related to health care (e.g. coverage monitoring surveys, access to education and for domestic or international travel).

In addition, a critical part of service delivery for PNC relies on empowering women, parents and families with information about their health. Digital interventions aimed at clients themselves – such as on-demand information services, targeted client communication (e.g. transmitting health information and reminders), reporting on health-system feedback (e.g. availability and quality of care) by clients, accessing their own medical records/personal health records/home-based records, and self-monitoring of their health and diagnostic data (20) – are all emerging approaches for complementing the services provided by health workers.

2

**Digital
adaptation kit
content for
postnatal care**



Component

Health interventions and recommendations

This DAK focuses on the following health interventions and recommendations related to PNC.

1.1 Interventions referenced in this DAK are based on the WHO universal health coverage (UHC) list of essential interventions

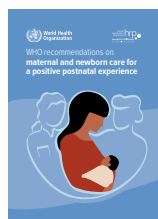
The key interventions for PNC are the following, as defined in the WHO UHC compendium of interventions, version 1.3 (25):

- » micronutrient supplementation for prevention of complications
- » support for healthy parenting
- » routine care for postpartum women
- » routine care for newborn/infant
- » support for healthy growth and development
- » screening and diagnosis of hearing impairment
- » vaccination based on individual characteristics.

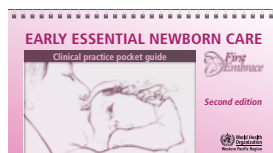
Additionally, as the WHO PNC recommendations complement existing WHO guidance covering interventions recommended throughout the maternal and newborn/infant health continuum, key interventions classified under categories that are not PNC but that are relevant to PNC, such as “immediate care of the newborn” (classified under the intervention category “labour and childbirth” in the WHO UHC compendium of interventions [25]) should also be added to the above list.

1.2 WHO guidelines, recommendations and guidance

The DAKs are intended to reflect health recommendations and content that has already been published in WHO guidelines and guidance documents. The health content and interventions for this DAK are based on the following WHO documents.



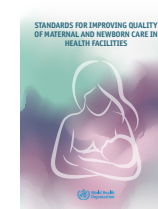
WHO recommendations on maternal and newborn care for a positive postnatal experience (2022) (19)



Early essential newborn care: clinical practice pocket guide, second edition (2022) (31)



Managing complications in pregnancy and childbirth: a guide for midwives and doctors, second edition (2017) (32)



Standards for improving quality of maternal and newborn care in health facilities (2016) (33)



Pregnancy, childbirth, postpartum and newborn care: a guide for essential practice, third edition (2015) (34)



Pocket book of hospital care for children, second edition (2013) (35)



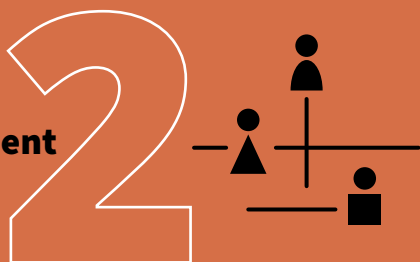
WHO recommendations: intrapartum care for a positive childbirth experience (2018) (36)

Other WHO guidance represented in the DAK includes:

- » WHO consolidated guidelines on tuberculosis – Module 2: screening (systematic screening for tuberculosis disease) (2021) (37)
- » WHO guideline on health workforce development, attraction, recruitment and retention in rural and remote areas (2021) (38)
- » WHO guidelines on physical activity and sedentary behaviour (2020) (39)
- » Classifying health workers: mapping occupations to the international standard classification (2019) (40)
- » Consolidated guidelines on HIV testing services 2019 (41)
- » WHO recommendations on home-based records for maternal, newborn and child health (2018) (42)
- » Guideline: protecting, promoting and supporting breastfeeding in facilities providing maternity and newborn services (2017) (43)
- » Guideline: preventive chemotherapy to control soil-transmitted helminth infections in at-risk population groups (2017) (44)
- » WHO recommendations on antenatal care for a positive pregnancy experience (2016) (45)
- » Medical eligibility criteria for contraceptive use, fifth edition (2015) (46)

- » [*WHO recommendations for prevention and treatment of maternal peripartum infections \(2015\) \(47\)*](#)
- » [*WHO recommendations on health promotion interventions for maternal and newborn health 2015 \(48\)*](#)
- » [*Ensuring human rights in the provision of contraceptive information and services: guidance and recommendations \(2014\) \(49\)*](#)
- » [*WHO recommendations: optimizing health worker roles to improve access to key maternal and newborn health interventions through task shifting \(2012\) \(50\)*](#)
- » [*Adolescent job aid: a handy desk reference tool for primary level health workers \(2010\) \(51\)*](#)

Component



Generic personas

A persona is a depiction of a relevant stakeholder, or “end-user”, of the system. Although the specific roles and demographic profiles of the personas will vary depending on the setting, the generic personas are based on the WHO core competencies and credentials of different health worker personas (52). Please note that these are developed from syntheses across multiple contexts as a starting point and further contextualization is required according to the needs, motivations and challenges of the targeted personas in each setting.

2.1 Targeted generic health worker personas

In the case of PNC, midwives, nurses, physicians and community health workers (CHWs) are the primary health worker personas for the digital client health record and decision-support system. The key competences of these health workers are defined by WHO ([Table 2](#)) (40).

Table 2. Descriptions of key generic health worker personas

Occupational title	Description	Different names	ISCO code
Midwifery associate professionals	Midwifery associate professionals provide basic health care and advice before, during and after pregnancy and childbirth. They provide advice to women, families and communities on birth and emergency plans, breastfeeding, newborn/infant care, family planning and related topics; monitor health status during pregnancy and childbirth; and implement care, treatment and referral plans usually established by medical, midwifery and other health workers.	Assistant midwife, auxiliary midwife (e.g. Bidan in Indonesia)	3222
Nursing associate professional	Nursing associate professionals provide basic nursing and personal care for people in need of such care due to effects of ageing, illness, injury, or other physical or mental impairment. They provide health advice to patients and families; monitor patients' conditions; and implement care, treatment and referral plans usually established by medical, nursing and other health workers.	Auxiliary nurse, nurse assistant, enrolled nurse (including definitions of nurse technician and associate nurse)	3221
Midwifery professional	Midwifery professionals plan, manage, provide and evaluate midwifery care services before, during and after pregnancy and childbirth. They provide birth care for reducing health risks to women and newborns according to the practice and standards of modern midwifery, working autonomously or in teams with other health workers. They may conduct research on midwifery practices and procedures and implement midwifery education activities in clinical and community settings.	Midwife, registered midwife, community midwife, professional midwife	2222

Occupational title	Description	Different names	ISCO code
Nurse professional	Nursing professionals provide treatment, support and care services for people who are in need of nursing care due to the effects of ageing, injury, illness, or other physical or mental impairment, or potential risks to health, according to the practice and standards of modern nursing. They assume responsibility for the planning and management of the patient care, including supervision of other health workers, working autonomously or in teams with medical doctors and others in the practical application of preventive and curative measures in clinical and community settings.	Registered nurse, nurse practitioner, clinical nurse specialist, advance practice nurse, practice nurse, licensed nurse, diploma nurse, BS nurse, nurse clinician, nursing professional, professional nurse	2221
Generalist medical doctor	Generalist medical doctors (including family and primary care doctors) diagnose, treat and prevent illness, disease, injury, and other physical and mental impairments, and maintain general health in people through application of the principles and procedures of modern medicine. They plan, supervise and evaluate the implementation of care and treatment plans by other health workers. They do not limit their practice to certain disease categories or methods of treatment and may assume responsibility for the provision of continuing and comprehensive medical care to individuals, families and communities.	Medical doctor (general), medical officer (general), physician (general), general practitioner, family medical practitioner, primary health care physician, district medical doctor, resident medical officer specializing in general practice, township medical officer, station medical officer, specialist in family medicine, medical officer	2211
Specialist medical doctor	Specialist medical doctors diagnose, treat and prevent illness, disease, injury, and other physical and mental impairments using specialized testing, diagnostic, medical, surgical, physical and psychiatric techniques, through application of the principles and procedures of modern medicine. They plan, supervise and evaluate the implementation of care and treatment plans by other health workers. They specialize in certain disease categories, types of patient or methods of treatment, and may conduct medical education and research activities in their chosen areas of specialization. Doctors in obstetric and gynaecological specialties and related branches focusing on the care of the reproductive system of women including before, during and after pregnancy and childbirth. Doctors in paediatrics and related specialties focusing on the prevention, diagnosis and treatment of health problems in infants, children and adolescents.	Specialist physician, gynaecologist, obstetrician, neonatologist, paediatrician, resident medical officer in specialist training	2212
Community health worker	Community health workers provide health education, referral, follow-up, case management, basic preventive health care and home visiting services to specific communities. They provide support and assistance to clients seeking family planning and to their families in navigating the health and social services system. For PNC, trained community health workers can conduct home visits during the first week after birth (19). The promotion of health-related behaviours for maternal and newborn/infant health^a by lay health workers is defined in WHO PNC recommendations (19).	Community health aide, health extension worker, community health promoter, village health worker, cadres	3253

ISCO: International Standard Classification of Occupations.

^a Including the promotion of the following: postnatal care, family planning (distribution of condoms [male and female] and other barrier methods, initiation and distribution of combined oral contraceptives, progestin-only oral contraceptives and emergency contraception, and information and general instructions on the Standard Days Method, Two Day Method and the lactational amenorrhoea method), postpartum HIV catch-up testing and retesting, sleeping under insecticide-treated nets, nutritional advice, nutritional supplements, basic newborn/infant care, exclusive breastfeeding and immunization according to national guidelines.

Source: Adapted from WHO (40).

2.2 Related personas

In addition to the targeted personas detailed above, there may be value in exploring other cadres and personas within the context of PNC services, such as social work and counselling professionals, pharmacists, and health service managers (see [Table 3](#)). However, these were not identified as the central personas for the data and decision-support content detailed in this DAK.

Table 3. Descriptions of related personas

Name	Description	Different names (if relevant)	ISCO code (if relevant)
Health-service user(s)	Postpartum women and the newborn/infant are the primary health-service users receiving PNC services from the targeted health worker personas (19,34). Postpartum woman: a woman in the period beginning immediately after childbirth and extending up to six weeks (42 days) after the birth Newborn: infant < 1 month of age Infant: a child > 1 month and < 1 year of age	Woman, postpartum woman, client Infant, newborn, neonate, child, client	N/A
Partner, parent, caregiver, family	A partner is the woman's spouse/husband/individual the woman is living with or in a close relationship with and/or the newborn's/infant's father. A parent is an individual responsible for caring for the newborn/infant (often this individual is the mother or the father, and may include single parents, coparents, same sex parents or parents with gender diverse identities). A caregiver coincides with the different individuals or family members who may be responsible for the care of the newborn/infant. Families are groups of individuals that play a supportive role for women and parents and may aid in caregiving responsibilities for the newborn/infant (19).	Partner, spouse, husband, parent, family member, caregiver	N/A
Social work and counselling professionals	Social work and counselling professionals provide counselling, therapy and mediation services to individuals, families, groups and communities in response to social and personal difficulties. They help clients to develop skills and access resources and support services needed to respond to issues arising from health problems, life transitions, addictions and other personal, family and social problems. They liaise with other social service agencies, educational institutions and health workers to advocate for client and community needs (40).	Social worker, counsellor, clinical social worker, district social welfare officer	2635
Pharmacist	Pharmacists store, preserve, compound and dispense medicinal products. They counsel on the proper use and adverse effects of drugs and medicines following prescriptions issued by medical doctors and other health workers. They contribute to researching, testing, preparing, prescribing and monitoring medicinal therapies for optimizing human health (40).	Chemist, clinical pharmacist, community pharmacist, hospital pharmacist, retail pharmacist, dispensing chemist	2262
Health service managers	Health service managers plan, direct, coordinate and evaluate the provision of clinical and community health services in hospitals, clinics, public health agencies and similar organizations. They provide overall direction, policy standards and operational criteria for the units they manage, including supervising and evaluating the recruitment, training and work activities of personnel. They monitor the use of health services and resources. They liaise with other health and welfare service providers, boards and funding bodies to coordinate the provision of services (40).	Health facility administrator, facility-in-charge, facility manager, district manager	1342

Name	Description	Different names (if relevant)	ISCO code (if relevant)
Clerical support worker	This group covers clerical support workers (excluding specialized health information technicians and medical secretaries) working in health systems including general clerks, keyboard clerks, client services clerks, material recording clerks and others who record, organize, store, compute and retrieve information, and perform clerical and secretarial duties in connection with money-handling operations, requests for information and appointments (40).	General office clerk, general secretary, general receptionist, accounts clerk, data entry clerk, hospital admissions clerk, human resources assistant, medical appointments schedule clerk, medical commodities stock controller	N/A

ISCO: International Standard Classification of Occupations; N/A: not applicable.

2.3 Additional considerations for contextualizing personas

Although this section provides an overview of the generic roles of the targeted personas, it is important to contextualize these personas to your setting. The generic personas described above can be supplemented by reflecting on these additional considerations.

- » **Background and demographics** (e.g. gender, age, whether they are from the community, familiarity with digital devices, do they own a mobile phone/smartphone?).
- » **Local environment** and any relevant contextual information about their surroundings (e.g. rural or urban; availability of electricity, water, internet).
- » **Expected roles and responsibilities:** What are the expected roles and responsibilities based on country context? How does this differ from the roles and responsibilities defined by WHO?
- » **Actual roles and responsibilities:** What are their actual roles and responsibilities, if there is any difference from what is expected?
- » **Health systems context:** How are they compensated? How far away is the nearest referral facility? What other personas/health workers do they interact with? Are there any digital systems used for health-service delivery?
- » **Challenges:** What day-to-day challenges might the end-user face?
- » **Motivations:** What does success look like to them? Are there targets they need to achieve?

For more details on persona development, please refer to the WHO *Digital transformation handbook for primary health care* (30).

Component 3 User scenarios



3.1 How to interpret user scenarios for functional requirements

User scenarios are helpful tools not only to better understand the context in which a digital tool would operate, but also for some insights into what key data elements would need to be recorded and accounted for in the database. Additionally, the context in which the tool would operate, illuminated by the user scenarios, provides insight into some functional and non-functional requirements that the system would also need. For example, highlighted in **yellow** are some key data elements that need to be recorded and/or calculated. Highlighted in **blue** is some decision-support logic that can be automated in the system. Highlighted in **green** are some key functional and non-functional requirements that should be included in the system and **bolded** are some adolescent-specific considerations that should be accounted for.

For example, the interpretation of the user scenario for first PNC contact (within 24 hours) and follow-up PNC contact is shown in [Table 4](#).

Table 4. Interpretation of the first PNC contact (within 24 hours) and follow-up PNC contact user scenario

Data element to be collected for the woman	Decision logic to be embedded	Functional and non-functional requirements
» Unique ID » Name » Date of birth » Marital status » History of chronic illnesses » Obstetric and past pregnancy history – information on childbirth » Vaccination history » Current medications » Risk factors » Services and medications provided	» Calculation of age based on date of birth » Decision-support logic to trigger certain pop-ups and reminders based on the time passed since childbirth	» Password protection » User management (login system) » Pop-up messages and reminders » Information icon for more detailed information and guidance » Recording data for later use

3.2 User scenario for first PNC contact (within first 24 hours) and follow-up PNC contact

Key personas

Mother: Pamela

Baby: Peter

Health workers: Joy (nurse), Patience (physician), Elena (midwife)

Pamela is 28 years old, is in good health and gave birth at 38 weeks to Peter, a healthy baby boy.

This was a planned pregnancy, she had eight antenatal care contacts and she had her first child vaginally three years ago without complications.

Pamela went into spontaneous labour at 38 weeks and went to the health facility identified in her birth plan.

On the labour ward, Peter was placed in skin-to-skin contact with his mother and breastfed in the first hour after birth. Both mother and baby were observed regularly for one hour after childbirth and then transferred together to the postnatal ward.

On the postnatal ward, Joy (nurse) easily finds Pamela's birth records and reviews Pamela's medical and obstetrical history as well as information on labour and childbirth; she then proceeds to ask for Pamela's consent to conduct a physical examination and, once obtained, she assesses Pamela, checking vaginal bleeding, uterine tonus, fundal height, temperature and heart rate, and then provides counselling on hygiene, perineal and breast care, breastfeeding, rest, postponing the first bath of the baby, and care for the baby. Joy records all findings from the clinical assessment and information on the provided counselling on a password-protected computer. Mother and baby are then monitored every six hours, with no evidence of abnormalities or danger signs.

The next day, Pamela and Peter are assessed clinically by Joy and Patience (physician). Prior to discharge, counselling on preventive measures (e.g. for prevention of postpartum mastitis, constipation, and infection, as well as schistosomiasis and anthelmintic treatment, and HIV prophylaxis), awareness of newborn danger signs and sleeping position for the prevention of sudden infant death syndrome (SIDS) is offered. Comprehensive information on family planning is also provided, with a detailed explanation of the available contraceptive measures in the postpartum period. In addition, Joy and Patience also assess Pamela's emotional well-being, her skills and confidence to take care of herself and of Peter as well as any factors in the home environment that may affect either the ability to provide care and/or care-seeking for herself and the baby. A home contact in two days is scheduled and the date of the follow-up postnatal care contact is recorded on the facility computer.

At 72 hours after birth, Elena (midwife) carries out a home visit. During the visit, Elena reviews birth data, performs an assessment of Pamela's physical and emotional well-being, of the family routine, and of Peter's growth and well-being. Pamela receives further counselling on breastfeeding, including advice on the management of common difficulties, recognizing and responding to the baby's cues for feeding, and information on the use and risks of feeding bottles, teats and pacifiers. After a complete clinical examination, Pamela is again advised on danger signs, home care, postpartum depression and anxiety, newborn care and family planning. Elena explains to Pamela the schedule for postnatal care contacts and advises her to attend the next contact in two weeks and then at six weeks postpartum. Elena logs all the information into her digital tool so that she can remember what counselling Pamela and Peter will need she next time she sees them.

Corresponding business processes (see [Component 4](#))

This scenario refers to the following business processes:

B. Routine first PNC contact (within first 24 hours)

D. PNC follow-up contact

3.3 User scenario for an adolescent client

Key personas

Mother: Angela
Baby: Camila
Health worker: Fernando (midwife), Adriana (nurse)

Angela is a 16-year-old teenager in her first pregnancy. The father of the baby is Angela's boyfriend, Alex. When Angela and Alex discovered the pregnancy, they were initially very fearful of the possible reaction of their families to the news. Once the pregnancy was revealed to their parents, both Angela and Alex received support, which helped them accept the pregnancy and prepare for the arrival of their baby.

Angela started antenatal care late, as she hid the news of the pregnancy for some time, fearing her parents' reaction. ANC follow-up was routine, except for a urinary tract infection at 24 weeks, which was adequately treated.

She gave birth at 39 weeks at the local maternity hospital without complications during labour or childbirth. She had a grade 1 perineal tear that did not require any suturing. A healthy baby girl, Camila, was born, and immediate skin-to-skin contact was facilitated as well as support to initiate breastfeeding as soon as possible. Angela and Camila were referred to rooming-in¹ after the 2-hour observation period following childbirth and monitored every 6 hours by Fernando (midwife), with no evidence of complications arising.

Angela and Alex receive guidance in the immediate postpartum from Fernando. Fernando reviews Angela's birth records on a password-protected tablet, then asks Angela a series of questions; after obtaining consent for physical examination, Fernando proceeds to the assessment of the mother, checking vaginal bleeding, uterine tonus, fundal height, temperature and heart rate. Fernando provides information on hygiene, breast care, rest, breastfeeding on demand, care for the newborn and postponing the baby's first bath, as well as information on family planning. Angela opts for a postpartum copper intrauterine device (IUD) as contraceptive method.

The next day, Fernando evaluates Angela and Camila. After a review of antenatal care and birth data, serologies, vital signs from the last 24 hours, and of the physical conditions and well-being of mother and baby, both are discharged. Prior to discharge, Angela receives further counselling on breastfeeding, including advice on the management of common difficulties, recognizing and responding to the baby's cues for feeding, and the use and risks of feeding bottles, teats and pacifiers. Angela's emotional well-being, her skills and confidence to take care of herself and of Camila (together with Alex) as well as any factors in the home environment that may affect either the ability to provide care and/or care-seeking for herself and the baby are also assessed. Fernando schedules a contact at 48 hours and records it on the facility computer.

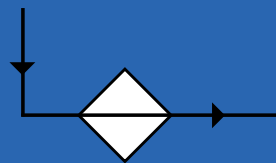
Angela attends the next postpartum evaluation at the primary health unit closest to her home, accompanied by Camila, Alex and her mother. During this contact, Adriana (nurse) reviews the clinical charts from the health facility and assesses the health and well-being of Angela and of Camila. After a complete clinical examination, provision of counselling on danger signs, care in the home, puerperal depression and anxiety, newborn care and breastfeeding is offered. Adriana explains to Angela the schedule for postnatal care contacts and advises her to attend the next contact in two weeks and then at six-weeks postpartum. Adriana logs all the information into her digital tool so that she can remember what counselling Angela will need she next time she sees her.

Corresponding business processes
(see [Component 4](#))

This scenario refers to the following business processes:
B. Routine first PNC contact (within first 24 hours)
D. PNC follow-up contact

¹ "Rooming-in" is a practice that involves keeping mothers and newborns together in the same room, immediately after leaving the labour and delivery room after a normal facility birth or from the time when the mother is able to respond to the newborn/infant, until discharge (43).

Component



Generic business processes and workflows

A business process, or process, is a set of related activities or tasks performed together to achieve the objectives of the health programme area, such as registration, counselling, referrals (1,26). Workflows are a visual representation of the progression of activities (tasks, events, interactions) that are performed within the business process (26). The workflow provides a “story” for the business process being diagrammed and is used to enhance communication and collaboration among users, stakeholders and engineers.

This DAK focuses on key business processes conducted by the personas (described in [Component 2](#)) as part of PNC service provision. These business processes are described in [Table 5](#). For each of these business processes, the corresponding business processes, data elements and decision-support needs are detailed within the following sections of this document.

Table 5. Overview of key PNC business processes

No.	Process name	Process ID	Personas	Objectives	Task set
	Title	ID used to reference this process throughout the DAK	Individuals interacting to complete the process	A concrete statement describing what the process seeks to achieve	The general set of activities performed within the process
A	Registration	PNC.A	» Woman » Newborn/infant » Health workers (one or more of: – CHW, – Nursing associate professional, or – Midwifery associate professional.	To identify and register or update registration information for a postpartum woman and/or newborn/infant so that she and/or the newborn/infant can proceed to the PNC (first or follow-up) contact.	<i>Starting point:</i> The woman and/or newborn/infant arrive at the postnatal ward, receive a home visit, arrive at the health facility or office. » Determine reason for the contact and assess for danger signs to ensure the woman and/or the newborn do not need urgent medical attention. Provide referral if necessary. » Register or update record from other modules (antenatal and/or intrapartum) with registration information with postpartum woman’s and/or newborn’s/infant’s details.

No.	Process name	Process ID	Personas	Objectives	Task set
B	PNC contact within first 24 hours	PNC.B	» Woman » Newborn » Health workers (one or more of: – CHW, – Nursing associate professional or midwifery associate professional, – Nurse, – Midwife, – Generalist medical doctor, – Specialist medical doctor, or – Social work professional).	To provide routine PNC services and counselling to the postpartum woman and newborn within the first 24 hours after birth.	<i>Starting point:</i> The woman and/or newborn have been registered and are ready to begin the first PNC contact in postnatal ward or at home. » Record profile and health history. » Query pregnancy health record and labour record/birth information. » Assess current postpartum mother and/or newborn conditions, including symptoms, and well-being, breastfeeding and medications. Conduct a physical exam, including vital signs, and psychological status. » Provide counselling on all relevant health topics for both woman and newborn (e.g. hygiene, diet, exercise and rest, newborn care [feeding, immunization and responsive care], danger signs, postpartum contraception, psychosocial and support services, risk reduction, diagnosis and treatment). » Distribute commodities as needed. » Schedule follow-up contact for the woman and/or the newborn. » Provide referral if necessary.
C	Emergency referral	PNC.C	» Woman » Newborn/ infant » Health workers (one or more of: – CHW, – Nursing associate professional or midwifery associate professional, – Nurse, – Midwife, – Generalist medical doctor, or – Specialist medical doctor.	To provide emergency referrals within a health facility, or to a higher-level health facility.	<i>Starting point:</i> The woman and/or newborn/infant require an emergency referral to or within a health facility or to higher-level health facility. » Stabilize the woman's or the newborn's/infant's condition. » If woman and/or newborn/infant can be treated within the same health facility, transfer internally. If not, discuss referral options, contact referral health facility and arrange emergency transport. » Liaise with next-level facility for care upon arrival. » Provide emergency care for the woman and/or the newborn/infant. » Ask and look for feedback report from the high-level assistance for continued primary health care.
D	PNC follow-up contact	PNC.D	» Woman » Newborn/infant » Health workers (one or more of): – CHW, – Nursing associate professional or midwifery associate professional, – Nurse, – Midwife, – Generalist medical doctor, – Specialist medical doctor, or – Social work professional.	To provide routine follow-up PNC services and counselling in health facility, office or at home.	<i>Starting point:</i> Woman and/or newborn/infant receive follow-up PNC contact at health facility, office or at home. » Review and update record from other contacts, if needed. » Assess current postpartum mother and/or newborn/infant symptoms, and well-being. Conduct a physical exam, including vital signs. Provide counselling on all relevant health topics for both woman and newborn/infant (e.g. hygiene, breastfeeding, diet, exercise and rest, newborn care [feeding, immunization and responsive care], danger signs, postpartum contraception, psychosocial and support services, risk reduction, diagnosis and treatment). » Distribute commodities as needed. » Schedule follow-up contact or close PNC and provide referral for regular primary health care for the woman and/or the infant, depending on postpartum contact schedule and individual needs. » Provide referral if necessary.

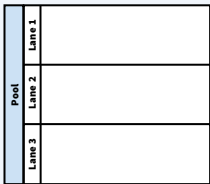
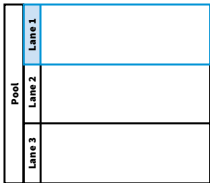
No.	Process name	Process ID	Personas	Objectives	Task set
E	Reporting on aggregate indicators	PNC.E	» Nursing associate professional or midwifery associate professional » Health service manager (at facility or district level)	Compile and submit relevant data registered in the digital system contributing to indicators and other reporting needs on a routine basis.	<i>Starting point:</i> Time for periodic (usually monthly) reporting. » Tally and compile required data (if on paper). » Check data quality. » Correct fixable errors. » Track loss to follow-up and monitor upcoming contacts for the women and newborns/infants in the catchment area. » Generate and review aggregate reports. » Submit to supervisor/director for approval. » Provide feedback and any changes required.

4.1 Overview of key processes

This section illustrates the workflows of the identified processes using standardized notations for business process mapping.

Table 6 first provides an overview of these notations.

Table 6. Business process symbols used in workflows

Symbol	Symbol name	Description
	Pool	A pool consists of multiple “swim lanes” that depict all the individuals or types of users that are involved in carrying out the business process or workflow. Diagrams should be clear, neat and easy for all viewers to understand the relationships across the different swim lanes. For example, a pool would depict the business process of conducting an outreach activity, which involves multiple stakeholders represented by different lanes in that pool.
	Swim lane	Each individual or type of user is assigned to a swim lane , a designated area for noting the activities performed or expected by that specific actor. For example, a PNC health worker may have one swim lane; the supervisor would have another swim lane; the clients would have another swim lane.

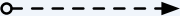
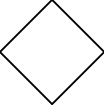
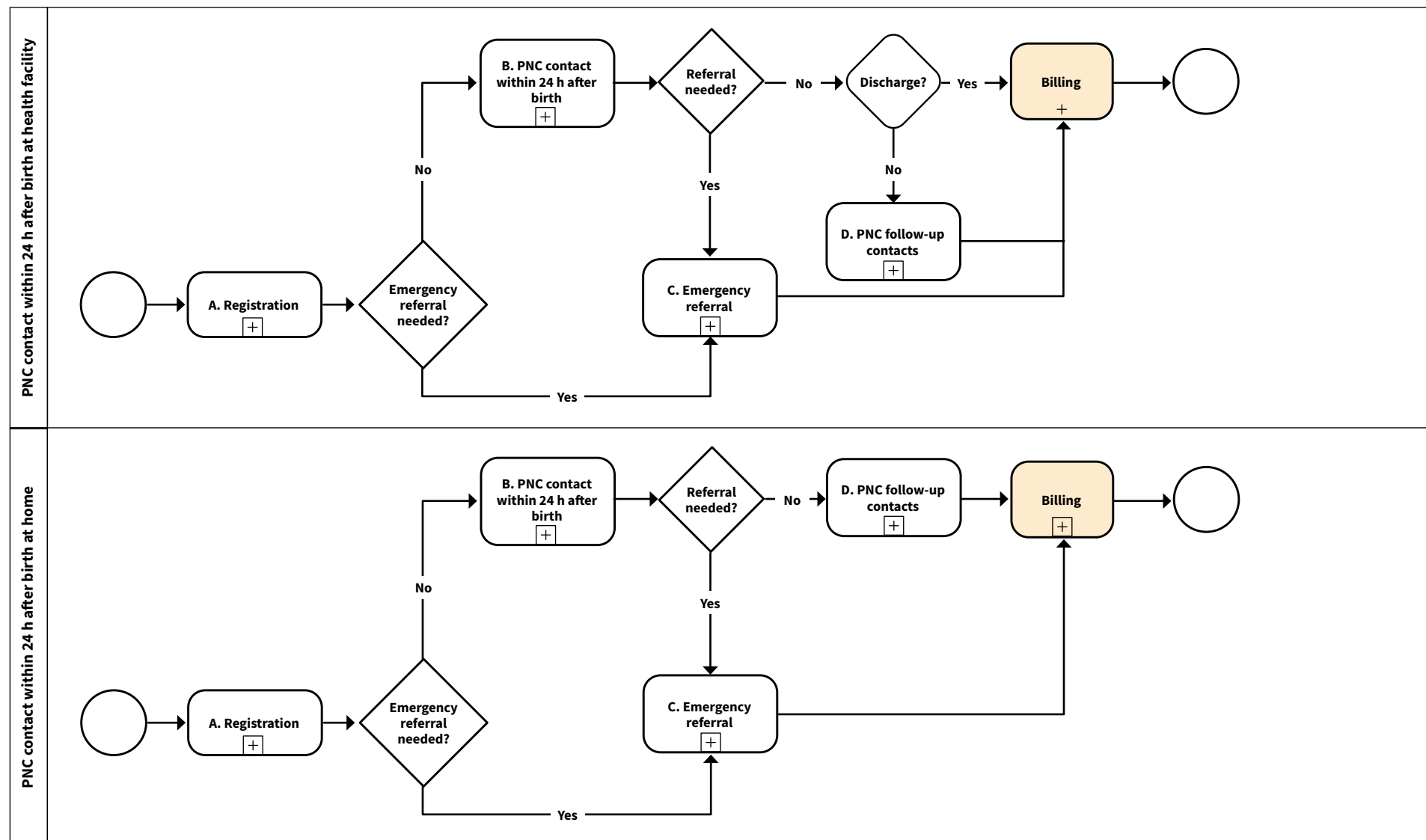
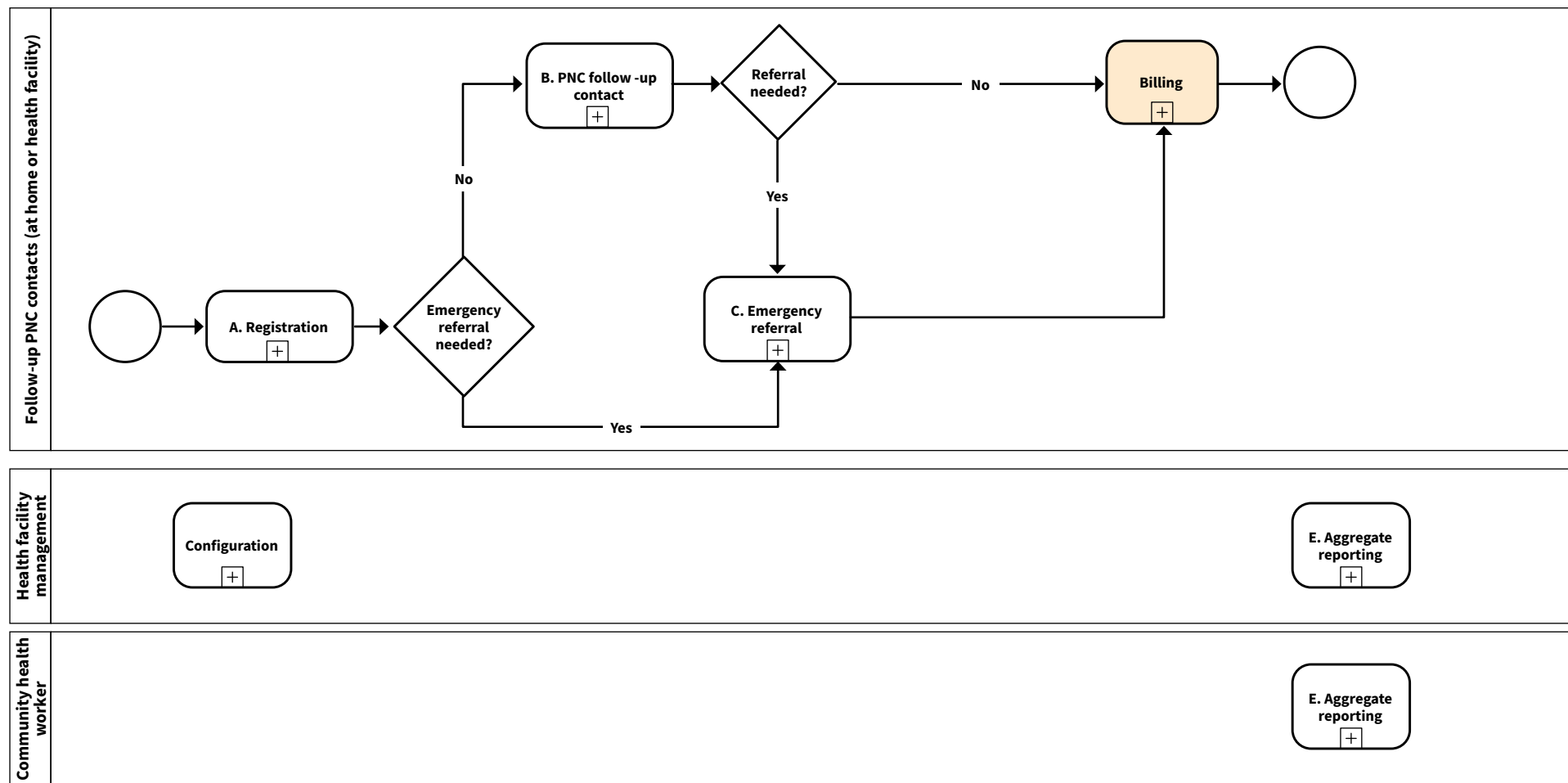
Symbol	Symbol name	Description
	Start event or trigger event	The workflow diagram should contain both a start and an end event , defining the beginning and completion of the task, respectively.
	End event	There can be multiple end events depicted across multiple swim lanes in a business process diagram. However, for diagram clarity, there should only be one end event per swim lane.
	Activity, process, step or task	Each activity should start with a verb, e.g. “Register client”, “Calculate risk”. Between the start and end of a task, there should be a series of activities noting the successive actions performed by the actor for that swim lane. There can also be subprocesses of each activity.
	Activity with subprocess	This denotes an activity that has a much longer subprocess to be detailed in another diagram. If the diagram starts to become too complex and unhelpful, the subprocess symbol should be used to reference another process depicted on another page.
	Activity with business rule	This denotes a decision-making activity that requires the business rule, or decision-support logic, to be detailed in a decision-support table. This means that the logic described in the decision-support table will come into play during this activity as outlined in the business process. This is usually reserved for complex decisions.
	Sequence flow	This denotes the flow direction from one process to the next. The end event should not have any output arrows. All symbols (except start event) may have an unlimited number of input arrows. All symbols (except end event and gateway) should have one and only one output arrow, leading to a new symbol, looping back to a previously used symbol or to the end event symbol. Connecting arrows should not intersect (cross) each other.
	Message flow	This denotes the flow of data or information from one process to another. This is usually used for when data are shared across swim lanes or stakeholder groups.
	Gateway	This symbol is used to depict a fork, or decision point, in the workflow, which may be a simple binary (e.g. yes/no) filter with two corresponding output arrows, or a different set of outputs. There should only be two different outputs that originate from the decision point. If you find yourself needing more than two “output” or sequence flow arrows, you most likely are trying to depict “decision-support logic” or a “business rule”. This should be depicted as an “Activity with business rule” (above) instead.
	Throw - Link	The “ Throw - Link ” serves as the start an off-page connector. It is the end of the process when there is no more room on your page for that workflow. It is the end of a process on your current page or the end of a subprocess that is part of a larger process. There will need to be a “Catch - Link” that follows the “Throw - Link”.
	Catch - Link	The “ Catch - Link ” serves as the end an off-page connector. It is the start of the new process on a different page from the “Throw - Link” or the start of a subprocess that is part of a larger process. There needs to be a “Throw - Link” that is aligned to the “Catch - Link”.
	Ad hoc subprocess	An ad hoc subprocess can contain multiple tasks. One or more tasks in this shape should be performed, and they can be performed in any order. However, not all of these activities need to be finished before moving on to the next activity.

Fig. 4. Overview of key PNC processes – health facility or home birth^a





PNC: postnatal care.

^a See [Table 6](#) for key.

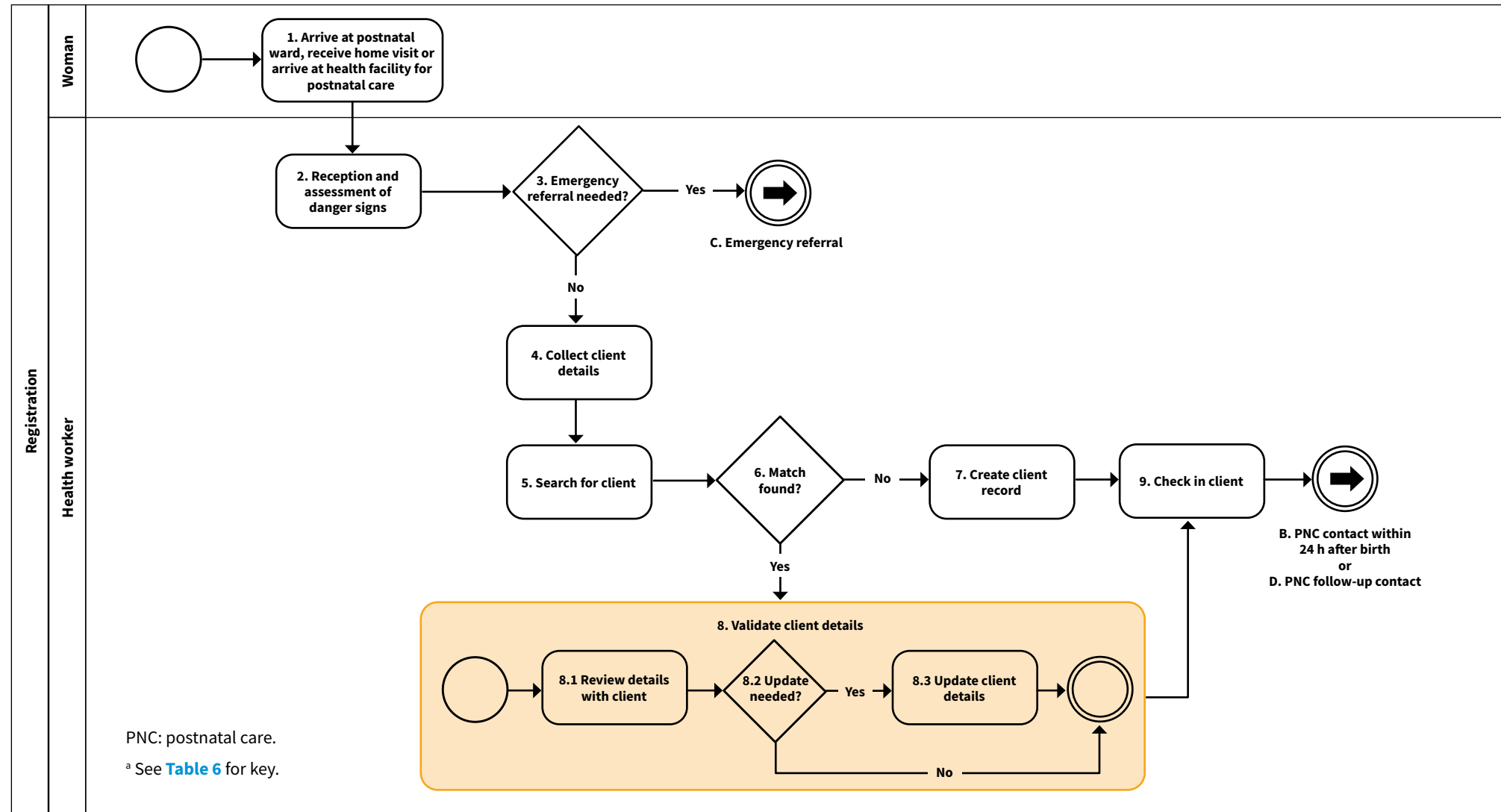
4.2 Workflows

For the purposes of this DAK, the term “client(s)” is used for general processes, such as registration, that may apply for other health areas; the appropriate term “woman”, “newborn” or “infant” is used in contexts of PNC-specific processes. The term “woman” refers to individuals who have given birth and includes those who identify as gender diverse individuals.

A. Business process for registration

Objective: To identify and register or update registration information for a postpartum woman and/or newborn/infant so that she and/or the newborn/infant can proceed to the PNC (within 24 hours after birth or follow-up) contact.

Fig. 5. Workflow A: Registration business process^a



Registration business process notes and annotations

Note: Registration activities can be performed either in the health facility or in the community (e.g. woman's home), depending on the context.

1. Arrive at health facility or client's location

- » Woman and/or newborn/infant arrive at the health facility or office.
- » Woman and/or newborn/infant could already be in the health facility (e.g. if the birth takes place at the facility).
- » Health worker arrives at the location of the woman and/or newborn/infant (e.g. woman's home in case of a home visit, e.g. following a home birth).

2. Reception and assessment of danger signs for the woman and/or newborn/infant

- » Assess the woman and/or newborn/infant for any possible visible danger signs.

3. Emergency referral necessary?

Determine whether emergency referral is required based on the assessment of danger signs, either from the client's location (e.g. home) to a facility or to a higher-level provider within the same facility or from a primary level facility to a higher-level referral facility.

Note: If yes, initiate referral (see Business process C: Emergency referral).

4. Collect client(s) details

- » Ask the woman whether she has previously been issued a unique identifier.
- » Ask the woman whether the newborn/infant has been issued a unique identifier.
- » Does the woman have a card/number/barcode?
- » Does the newborn/infant have a card/number/barcode?
- » Determine whether the woman is new to the health facility/health post/catchment area in the community.
- » For returning clients, details will be retrieved from the registry of clients at this facility or, if possible, from a central client registry.

5. Search for client(s)

- » This search process can be done through a variety of means depending on what mechanisms are available in country. For example, clients can be searched for by using their name, unique identifier, a QR code or even biometrics.

6. Match found?

- » If multiple records are found with no unique ID, provide option to merge records.

7. Create client(s) record

- » Issue a unique identifier if used and possible at the facility.

8. Validate client(s) details

- » Review and update woman's and/or newborn's/infant's record.

8.1. Review data found in the system with client(s)

- Review woman's and/or newborn's/infant's non-clinical information – name, address, date of birth, contact information (for woman only), etc.

8.2. Update needed?

- Has the woman moved? Has she changed contact information or has any other information changed?

8.3. Update client details

- Woman can provide updated information if she has moved or changed her details recently.
- Merge/update client records.

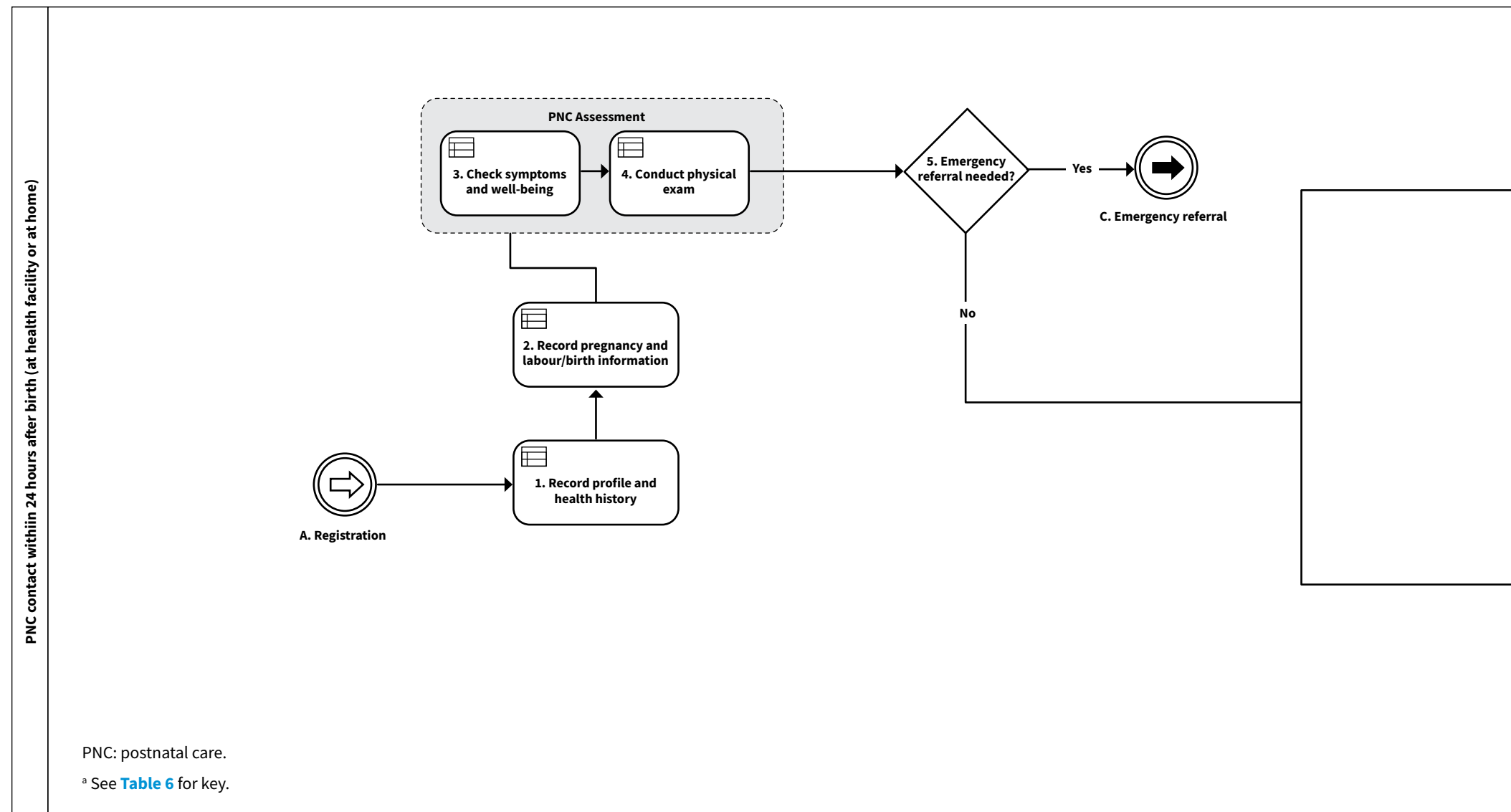
9. Check in the client(s)

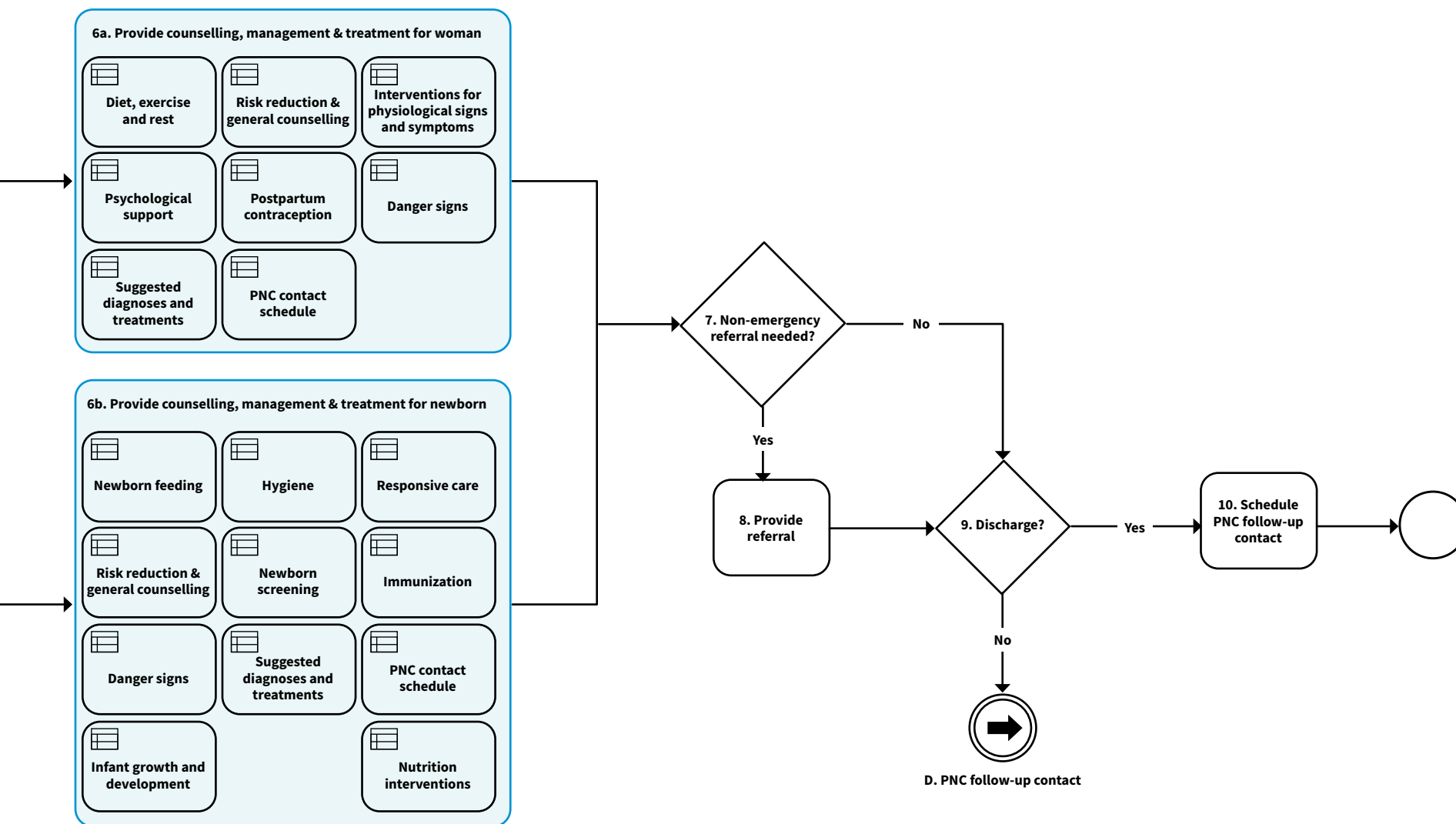
- » Record woman's and/or newborn's/infant's updated details in the client registry.
- » Add woman and/or newborn/infant to the relevant queue for PNC services if at health facility.
- » Initiate PNC services if at home.
- » Send/share intake confirmation to/with referring facility as warranted.

B. Business process for PNC contact within first 24 hours

Objective: To provide routine PNC services and counselling to the postpartum woman and newborn within the first 24 hours after birth.

Fig. 6. Workflow B: PNC contact within first 24 hours business process^a





PNC contact within 24 hours after birth business process notes and annotations

A. Registration

- » The registration process is to record administrative and demographic information as detailed in Business process A: Registration.

Note: If emergency referral is needed, health worker refers immediately based on danger signs presented and proceeds to Business process C: Referral.

1. Record profile and health history

- » Record information on past medical and obstetric history, medications, health-related behaviour, immunization status and other background information for the woman and/or newborn.

2. Record pregnancy and labour/birth information

- » Record information on past pregnancy and additional obstetric history, ANC and on labour and birth.

Note: In some digital contexts the information on ANC may be extracted from the ANC and/or the labour and birth record.

Note: Steps 3–6 may occur in parallel or in a different order. Not all steps may need to be completed.

3. Check symptoms and well-being

- » Assess the presence or absence of any symptoms or specific complaints.
- » Assess the woman's and/or newborn's well-being and health-related behaviours.

4. Conduct physical exam

- » Conduct a medical examination of the woman and/or newborn.
- » Record vital signs (e.g. blood pressure, heart rate, respiratory rate, weight, temperature) and all relevant exam results.

5. Emergency referral needed?

- » Determine whether emergency referral is required based on the PNC assessment, whether to a higher-level provider within the same facility, a higher-level referral facility or from home to a health facility.

Note: If yes, initiate referral (see Business process C: Emergency referral).

6a and 6b. Provide counselling, management and treatment for the woman and/or the newborn

- » Based on the previous steps, provide counselling for diet, exercise and rest, newborn feeding (including exclusive breastfeeding), hygiene, potential risk behaviours and danger signs, as well as preventive services (i.e. immunization), infant growth and development, the PNC contact schedule, and diagnoses and any treatment that can be provided at home or the health facility.

Note: Newborn screening for eye red reflex and hearing impairment with otoacoustic emissions (OAE) require specialized equipment and training and are usually done at facility level. If contact is at home, non-emergency referral can be made for follow up.

7. Non-emergency referral needed?

- » If there are any diagnoses requiring referral or services that cannot be provided at health facility, office or at home, the health worker refers postpartum woman and/or newborn based on findings during PNC assessment, counselling and management steps. These services could also include psychosocial, emotional, cultural and social support services to help the postpartum woman and/or newborn. Examples of these services are psychosocial counselling, peer-support organizations, breastfeeding support, and social service organizations for financial assistance, nutrition assistance and education, and childcare.

8. Provide non-emergency referral

- » Provide information on referral facilities/outpatient services and referral slip for sharing information with referral staff, as necessary. If possible, specifically mention:
 - why referral is needed.
 - what transport options are available (including location and directions on how to get to the referral facility/outpatient service).
 - who to see and what is likely to happen.
 - follow-up on return, if needed.

Note: If there are various possible referral sites that provide the required level of care, either the woman or the newborn's/infant's caregiver should indicate their preferred referral location.

9. Discharge (only for health-facility birth)

- » Conduct a discharge assessment of the woman and/or newborn.

Note: If either one cannot be discharged, proceed to follow-up contact (see Business process D: PNC follow-up contact).

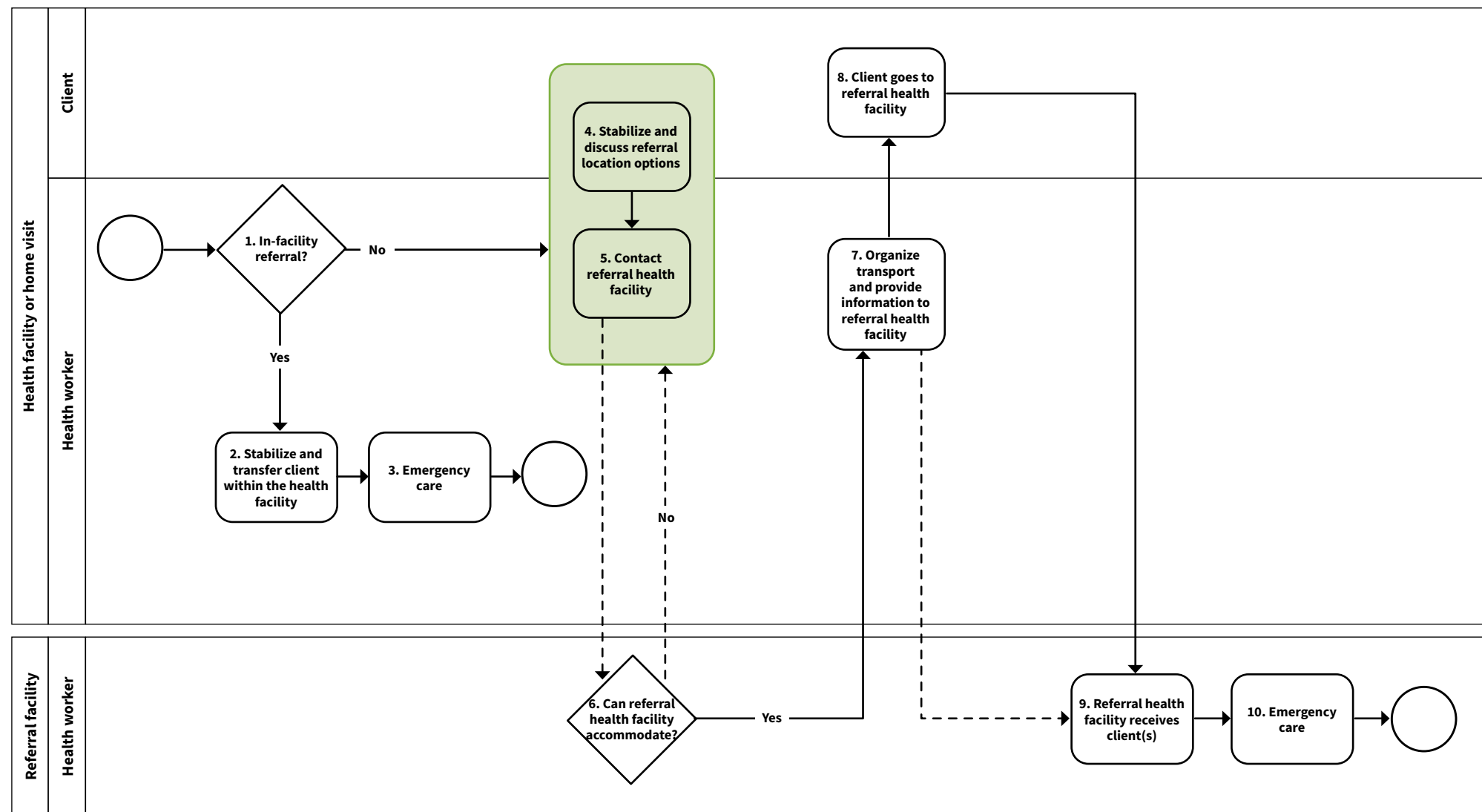
10. Schedule PNC follow-up contact

- » Schedule a PNC follow-up contact.

C. Business process for emergency referral

Objective: To provide timely and appropriate referrals to or within a health facility, or to a higher-level health facility.

Fig. 7. Workflow C: Emergency referral business process^a



^a See Table 6 for key.

Referral business process notes and annotations

Note: If the woman and/or newborn/infant need an immediate referral due to an emergency, a referral can be made at any time including during registration and service provision workflows. The woman and/or newborn/infant are assumed to need emergency referral if their condition requires immediate medical attention.

Note: If the referral is not an emergency referral, the health worker will provide it after the standard steps in PNC first or follow-up contact process (see Business process B: PNC contact within 24 h after birth and/or D: PNC follow-up contact).

1. In-facility referral?

- » If the woman and/or newborn/infant are at a health facility, determine whether they can be treated by a higher-level provider or in a higher-level ward/care department.

2. Stabilize and transfer the client within the health facility

- » Stabilize woman and/or newborn/infant and give pre-transfer treatment as needed.
- » If the woman and/or newborn/infant is/are stable enough, transfer to higher-level ward/care department. If the woman and/or the newborn/infant is/are still not stable, provide prereferral treatment for stabilization, if possible. If unable to stabilize adequately for transfer, discuss with higher-level ward/care department staff.

3. Emergency care

- » Provision of necessary care.

4. Stabilize and discuss referral location options

- » Stabilize woman and/or newborn/infant and give prereferral treatment as needed.
- » Once stabilized, in discussion with the woman/her relatives/the newborn's/infant's carer, decide on where the woman and/or newborn/infant will be referred. If possible, specifically mention:
 - why referral is needed.
 - what transfer options are available (including transportation options, location and directions on how to get to the referral facility).
 - who to see and what is likely to happen.
 - follow-up on return, if needed.
- » If there are various possible referral health facilities that provide the required level of care, either the woman or her relatives or the newborn's/infant's carer should indicate their preferred referral location.

5. Contact referral health facility

- » Contact the referral facility to determine whether that facility can accommodate such a referral and to ensure that the woman and/or newborn/infant are treated as soon as possible upon arrival.

6. Can referral health facility accommodate?

- » Check whether the referral health facility can accommodate the woman and/or newborn/infant and provide the services needed. If the referral facility can accommodate the woman and/or newborn/infant, move on to step 7. Otherwise, find a different facility that can accommodate the woman and/or newborn/infant.

Note: A system can be set up to validate that a facility has the supplies, skills and capacity to accommodate the woman and/or newborn/infant.

7. Organize transport and provide information to referral facility

- » Referring health facility arranges transport – usually by phoning for an ambulance or other vehicle.
- » Documentation should be sent with the patient for review by the receiving health workers on the woman's/newborn's/infant's arrival. Fill out a referral form, which can include notification of the referral destination.
- » Provide the necessary clinical, sociodemographic and identity information to the referral facility (note: this can be done digitally if the appropriate systems are in place).

8. Client(s) go to referral health facility

- » Client(s) proceeds to the referral facility.

9. Referral health facility receives client(s)

- » Referral facility receives the woman and/or newborn/infant, along with all the necessary clinical, sociodemographic and identification information.

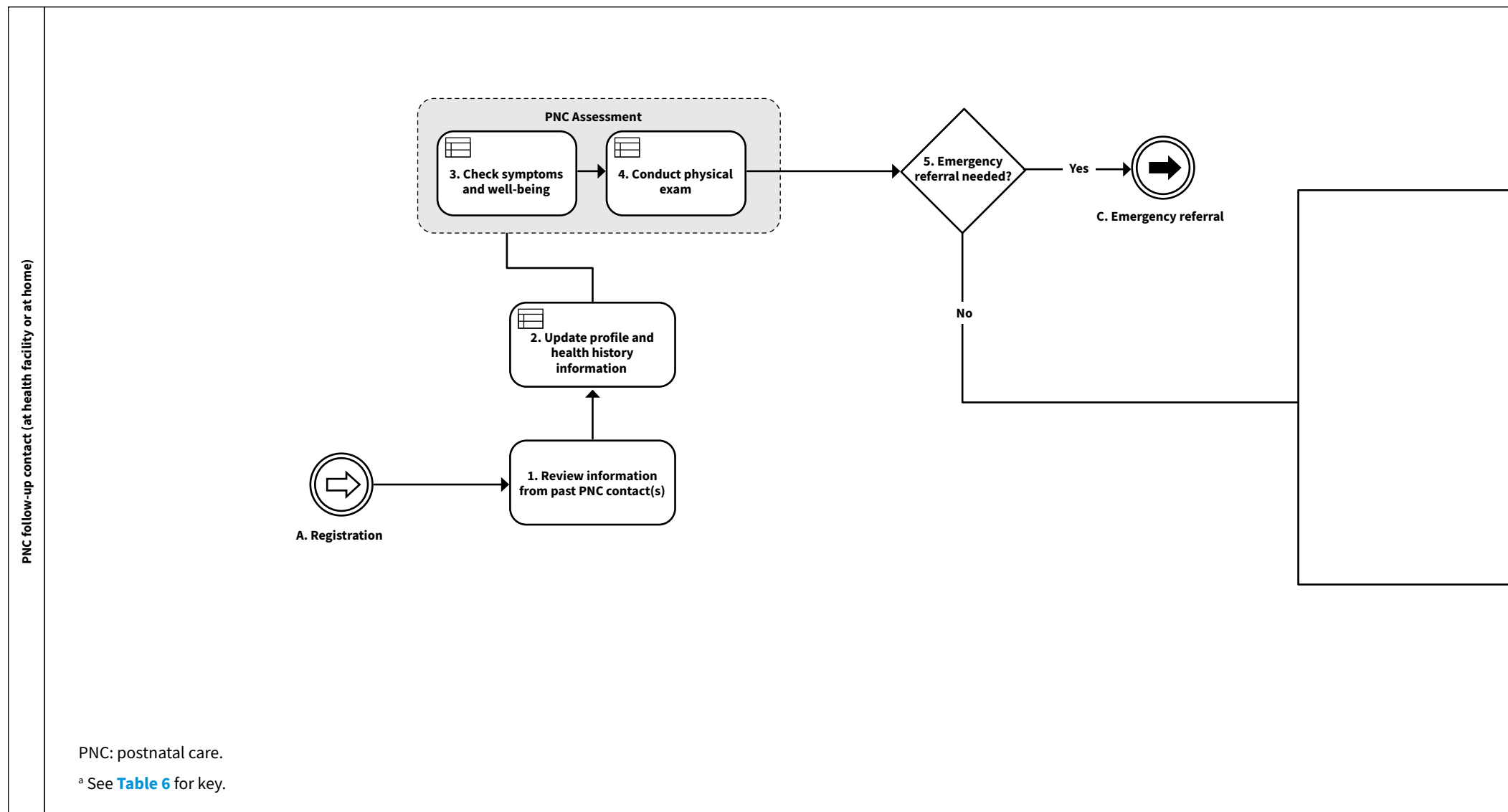
10. Emergency care

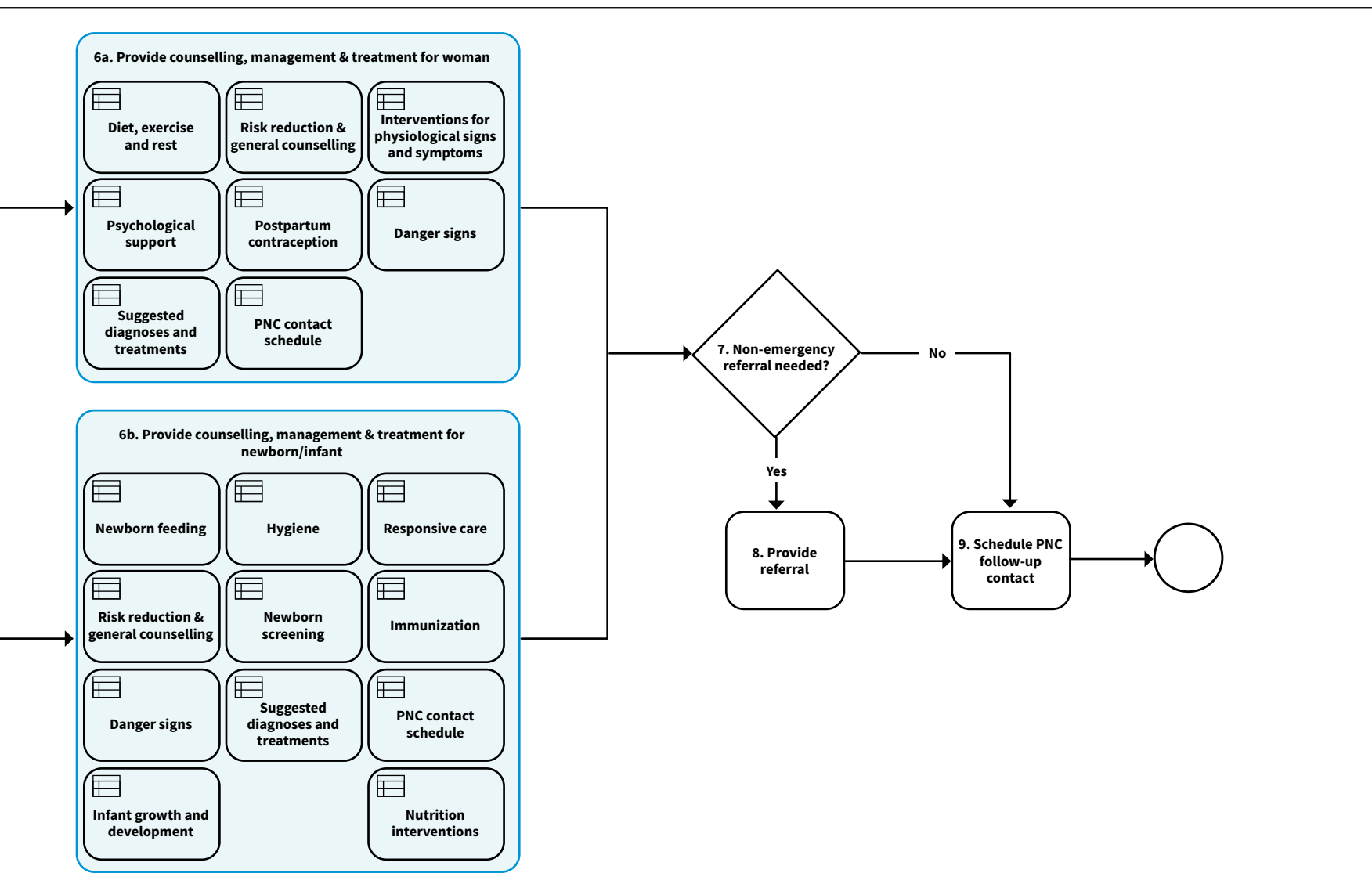
- » Provision of necessary care.

D. Business process for PNC follow-up contact

Objective: To provide routine follow-up PNC services and counselling at home, in the community as outpatient or at a health facility.

Fig. 8. Workflow D: PNC follow-up contact^a





PNC follow-up contact business process notes and annotations

A. Registration

The registration process is to record or update administrative and demographic information as detailed in Business process A: Registration.

Note: If emergency referral is needed, health worker refers immediately based on danger signs presented and proceeds to Business process C: PNC Emergency referral.

1. Review information from past PNC contact(s)

- » Review information for the woman and/or newborn/infant from past contact(s), as needed.

2. Update profile and health history information

- » Update any necessary information on past medical and obstetric history, medications, health-related behaviour, immunization status and other background information for the woman and/or newborn.
- » Follow-up on return information from emergency and non-emergency referrals from past contact(s).

Note: Steps 3–6 may occur in parallel or in a different order. Not all steps may need to be completed.

3. Check symptoms and well-being

- » Assess the presence or absence of any symptoms or specific complaints.
- » Assess the woman's and/or newborn's/infant's well-being and health-related behaviours.

4. Conduct physical exam

- » Conduct a medical examination of the woman and/or newborn/infant.
- » Record vital signs (e.g. blood pressure, heart rate, respiratory rate, weight, temperature) and all relevant exam results.

5. Emergency referral needed?

- » Determine whether emergency referral is required based on the PNC assessment, either from home to a health facility, to a higher-level provider within the same facility or a higher-level referral facility.

Note: If yes, initiate referral (see Business process C: Referral).

6a and 6b. Provide counselling, management and treatment for the woman and/or the newborn

- » Based on the previous steps and information recorded from past contact(s), health worker provides counselling for diet, exercise and rest, newborn/infant feeding (including exclusive breastfeeding), hygiene, potential risk behaviours and danger signs, as well as preventive services (e.g. immunization), infant growth and development, the PNC contact schedule, and diagnoses and any treatment that can be provided at home or the health facility.

Note: Newborn screening for eye red reflex and hearing impairment with otoacoustic emissions (OAE) require specialized equipment and training and are usually done at facility level. If contact is at home, non-emergency referral can be made for follow up.

7. Non-emergency referral needed?

- » If there are any diagnoses requiring referral or services that cannot be provided at home, health facility or outpatient services, the health worker refers postpartum woman and/or newborn/infant based on findings during PNC assessment, counselling and management steps. These services could also include psychosocial, emotional, cultural and social support services to help the postpartum woman and/or newborn/infant. Examples of these services are psychosocial counselling, peer-support organizations, breastfeeding support, and social service organizations for financial assistance, nutrition assistance and education, and childcare.
- » Provide information on referral facilities/outpatient services and referral slip, as necessary.

8. Provide non-emergency referral

- » Provide information on referral facilities/outpatient services and referral slip for sharing information with referral facility staff, as necessary. If possible, specifically mention:
 - why referral is needed.
 - who to see and what is likely to happen.
 - what transport options are available (including location and directions on how to get to the referral facility/outpatient service).
 - follow-up on return, if needed.

Note: If there are various possible referral health facilities that provide the needed non-emergency care, either the woman or the newborn's/infant's carer should indicate their preferred referral location.

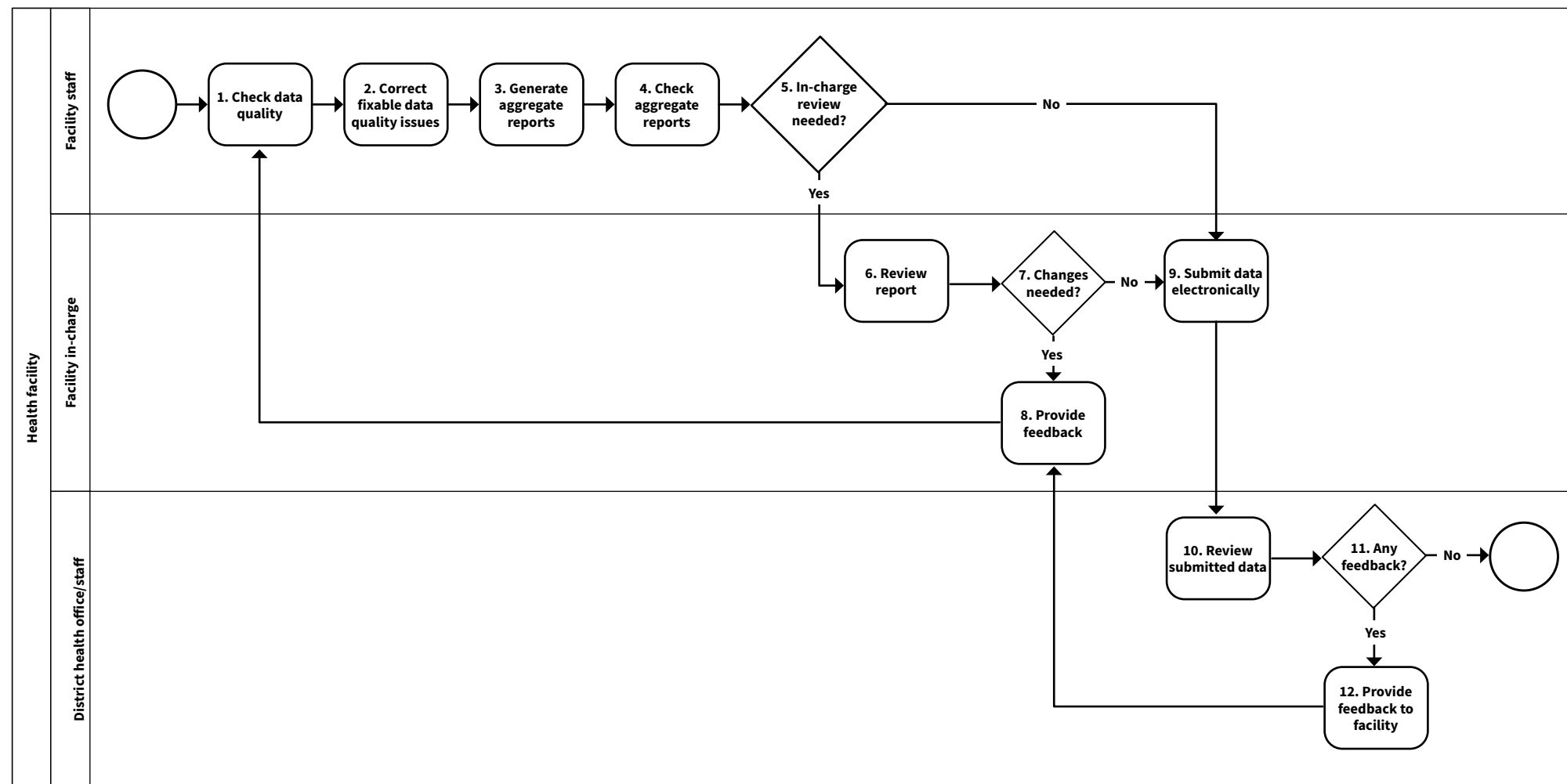
9. Schedule PNC follow-up contact

- » Schedule follow-up contact or close PNC and provide referral for regular primary health care for the well-woman and/or healthy child check-ups, depending on postpartum contact schedule and individual needs.

E. Business process reporting on aggregate indicators

Objective: Compile and submit relevant data registered in the digital system contributing to indicators and other reporting needs on a routine basis.

Fig. 9. Workflow E: Reporting on aggregate indicators^a



^a See Table 6 for key.

Reporting on aggregate indicators business process notes and annotations

1. **Check data quality**
 - » Health facility reviews accuracy, validity and completeness of data in system.
2. **Correct fixable data quality issues**
 - » Depending on local policy, this step might not be required.
3. **Generate aggregate reports**
 - » This can be automated and done digitally.
4. **Check aggregate reports**
 - » Depending on local policy, this step might not be required.
5. **Facility supervisor/in-charge review needed?**
 - » Determine whether the report needs to be reviewed by the person in charge of the health facility.
6. **Review report**
 - » The reports are reviewed by the facility supervisor/in-charge.
7. **Changes needed?**
 - » Facility supervisor/in-charge determines whether the report is accurate or has any issues.
8. **Provide feedback**
 - » If there is any issue with the report, the facility supervisor/in-charge will provide feedback to the responsible person to make corrections to client-level data.
9. **Submit data electronically**
 - » Submission can be done automatically or manually.
 - » Depending on the local policies, an active “submission” may not be needed, and the district-, provincial- and national-level MOH would be able to access data directly for reporting purposes.
10. **Review submitted data**
 - » District- or higher-level reviewer uses data in report to review progress and make decisions on improvements and actions to take.
 - » Reports and data may be used by the facility supervisor/in-charge or staff at multiple points during the business process, such as here, earlier in the process or outside of the business process.
11. **Any feedback?**
 - » District- or higher-level reviewer determines whether there is any feedback after review of the data.
12. **Feedback provided to the health facility**
 - » If there is any feedback, district- or higher-level reviewer will provide feedback to the health facility.

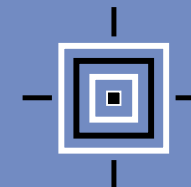
4.3 Additional considerations for adapting workflows

As a reminder, these workflows are meant to be generic and high level. They will require a level of customization and adaptation as they are translated into a digital system for a specific context. These workflows are considered to be “80% complete”, thus the “other 20%” will need to be done through a series of human-centred design methods and mechanisms to complete the workflows for an implementation. For example, there might be additional workflows that need to be drawn out or there might be additional activities expected of a health worker in the facility. Some workflows are not included due to the high level of contextualization required, including billing, dispensing (if separate from service provision), configuration (of facility-level specifics) and follow-up (which can be automated). Alternatively, there might be some activities and tasks a health worker would not be expected to do. Although these workflows can be considered as a starting point, it is helpful to conduct further validation through interviews with the targeted health worker personas or shadowing their work to obtain a better sense of the differences that would need to be reflected in the digital system.

Component

5

Core data elements



This section outlines the minimum set of data corresponding to different points of the workflow within the identified business processes. This data set can be used on any appropriate software system and lists the data elements relevant for service delivery and executing decision-support logic, and for populating indicators and performance metrics. Although this section provides a high-level overview of the data elements, a more complete data dictionary in spreadsheet form detailing the input options, validation checks and concept dictionary codes is available [here](#).

Inclusion of a data element in the table does not by itself indicate that collection of those data is required. Additionally, some data elements are dependent on other data elements (e.g. test results are only entered when a test has been performed). Collection of data should not affect clinical care. This will require review and adaptation.

5.1 Simplified list of core data elements

Tables 7 and 8 provide a simplified list of core data elements – a snapshot of the comprehensive [data dictionary](#). As with the workflows, we view this data dictionary as “80% generic”, with the expectation that the “other 20%” will be supplemented and modified through country adaptation.

Additionally, please note that while some of these activities should be administered immediately after birth (e.g. vitamin K) and could be captured during intrapartum care, having this information within PNC data collection elements is necessary for continuity of care.

Table 7. Workflow core data elements for identified PNC business processes – maternal care

Activity ID. Activity name	Data element ID	Data element label	Description and definition
Business process PNC.A: Registration			
PNC.A1. Arrive at postnatal ward, receive home visit or arrive at health facility for postnatal care	N/A – No data are recorded during this activity		

Activity ID. Activity name	Data element ID	Data element label	Description and definition
PNC.A.2 Reception and assessment of danger signs	PNC.A2.DE.1–51	Reason for the PNC contact	Records the reason for the PNC contact today
	PNC.A2.DE.52–56	Individual accompanying the newborn/infant	Records the individual accompanying the newborn/infant
	PNC.A2.DE.57–59	Who will be evaluated now?	Record information on current patient (woman and/or newborn/infant)
	PNC.A2.DE.60–73	Woman’s danger signs	Before each contact, the health worker should check whether the woman has any of the danger signs listed here – if yes, they should refer to a facility (if contact is at home) or higher level of care; if no, they should continue to the normal contact
	PNC.A2.DE.74–91	Newborn’s/infant’s danger signs	Before each contact, the health worker should check whether the newborn/infant has any of the danger signs listed here – if yes, they should refer to a facility (if contact is at home) or higher level of care; if no, they should continue to the normal contact
PNC.A3. Emergency referral needed?	See PNC.C: Emergency referral		
PNC.A4. Collect client details	PNC.A4.DE.1–8	Woman’s unique identification	Unique identifier generated for new clients or a universal ID, if used in the country
	PNC.A4.DE.9–14	Newborn’s/infant’s unique identification	Unique identifier generated for new clients or a universal ID, if used in the country
	PNC.A4.DE.15–19	Person responsible for the newborn’s/infant’s identification	Obtain details of the person responsible for the newborn’s/infant’s identification
	PNC.A4.DE.20–24	Additional information	Additional information for alternative contact (e.g. partner, mother, sibling of the woman or person responsible for newborn’s/infant’s care)
	PNC.A4.DE.25–29	Payment	How will the client pay for health services and commodities?
PNC.A5. Search for clients	N/A – No data are recorded during this activity		
PNC.A6. Match found	N/A – No data are recorded during this activity		
PNC.A7. Create client record	See PNC.A4. Collect client details		
PNC.A8. Validate client details	See PNC.A4. Collect client details		
PNC.A9. Check-in client	N/A – No data are recorded during this activity		
Business process PNC.B: PNC contact within 24 hours after birth			
PNC.B1. Record profile and health history OR PNC.D2. Update profile and health history information	PNC.B1.DE.1–6	Highest level of education achieved	The highest level of schooling the woman has reached
	PNC.B1.DE.7–13	Occupation	The woman’s occupation
	PNC.B1.DE.14–24	Allergies	Does the woman have any allergies?
	PNC.B1.DE.25–36	Past surgeries	Has the woman had any prior surgeries?
	PNC.B1.DE.37–58	Existing chronic health conditions	Does the woman have any current chronic health conditions or problems?

Activity ID. Activity name	Data element ID	Data element label	Description and definition
	PNC.B1.DE.59–63	Tetanus toxoid-containing vaccine (TTCV) immunization history	The woman's history of receiving TTCV
	PNC.B1.DE.64–67	Flu immunization history	Whether or not this year's seasonal flu vaccine has been provided
	PNC.B1.DE.68–71	Coronavirus disease (COVID-19) immunization history	Whether or not the COVID-19 vaccine has been provided
	PNC.B1.DE.72–80	Clinical inquiry for alcohol and other substance use	Whether or not the health worker has performed a clinical enquiry into alcohol and substance use
	PNC.B1.DE.81–84	Tobacco use	Whether or not the woman uses tobacco products
	PNC.B1.DE.85–88	Condom use for prevention of Zika, HIV and other sexually transmitted infections	Whether or not the woman/her partner use condoms
PNC.B2. Record pregnancy, labour and birth information	PNC.B2.DE.1–10	Woman's obstetric history	Indicate the woman's obstetric history
	PNC.B2.DE.11–28	Past pregnancy complications	Indicate whether the woman has had any complications or problems in any previous pregnancy
	PNC.B2.DE.29–34	Substance use during pregnancy specification	If selected substance use, specify and indicate what type of substances used during past pregnancies
	PNC.B2.DE.35–38	Antenatal care (ANC)	Information on the ANC contacts
	PNC.B2.DE.39–42	Pregnancy planning	Information on pregnancy planning
	PNC.B2.DE.43–46	Pregnancy acceptance	Information on pregnancy acceptance
	PNC.B2.DE.47–61	ANC complications	Indicate whether the woman has had any complications or problems in this pregnancy
	PNC.B2.DE.62–65	Syphilis treatment	If selected syphilis, specify the specific treatment
	PNC.B2.DE.66–69	HIV testing at early pregnancy and retesting in the third trimester	Indicate whether the woman underwent HIV testing during pregnancy
	PNC.B2.DE.70–73	HIV treatment	If HIV positive in pregnancy, specify the specific treatment
	PNC.B2.DE.74–78	HIV catch-up testing at admission for labour	Indicate whether the woman underwent HIV rapid test at admission for labour
	PNC.B2.DE.79–84	Woman's blood type ABO	Specify the woman's blood type ABO
	PNC.B2.DE.85–89	Woman's blood type Rh	Specify the woman's blood type Rh
	PNC.B2.DE.90–96	Childbirth details	Specify details regarding childbirth
	PNC.B2.DE.97–100	Fever during labour or childbirth	Indicate whether the woman suffered from a fever during labour or childbirth
	PNC.B2.DE.101–104	Intrapartum prophylaxis for neonatal sepsis	Indicate whether the woman received prophylaxis for neonatal sepsis
	PNC.B2.DE.105–109	Amniotic fluid colour	Specify the amniotic fluid colour during labour

Activity ID. Activity name	Data element ID	Data element label	Description and definition
	PNC.B2.DE.110–114	Fetal presentation at birth	Specify the details of the foetal presentation at birth
	PNC.B2.DE.115–120	Mode of birth	Specify the details of the type of birth
	PNC.B2.DE.121–132	Labour and/or birth complications	Indicate whether the woman has had any complications or problems with labour and/or birth
	PNC.B2.DE.133–138	Perineal laceration	Specify the degree of perineal laceration
	PNC.B2.DE.139–142	Repair of perineal laceration	Indicate whether the perineal lacerations were repaired
	PNC.B2.DE.143–146	Newborn sex	Specify the phenotypic sex of the newborn
	PNC.B2.DE.147–149	Newborn status at birth	Specify the vital status of the newborn at birth
	PNC.B2.DE.150–157	Newborn measurements	Specify the newborn measurements at birth
	PNC.B2.DE.158–161	Umbilical cord clamping ≥ 1 minute after birth	Indicate whether umbilical cord clamping performed ≥ 1 minute after birth
	PNC.B2.DE.162–166	Skin-to-skin contact within one hour of birth	Indicate whether skin-to-skin contact was initiated in the first hour after birth
	PNC.B2.DE.167–170	Breastfeeding initiated within one hour of birth	Indicate whether the newborn was breastfed/put to the breast within one hour of birth
	PNC.B2.DE.171–174	Was the newborn resuscitated with bag and mask at birth?	Indicate whether the newborn required resuscitation with bag and mask at birth
	PNC.B2.DE.175–181	Intrauterine device (IUD) placement at childbirth	Indicate whether the woman had IUD placement at childbirth
	PNC.B2.DE.182–185	Female sterilization at childbirth	Indicate whether the woman had a tubal ligation at childbirth
PNC.B3a. Check symptoms and well-being within 24 hours after birth (maternal)	PNC.B3a.DE.1–5	Screening for postnatal depression and anxiety	Specify the results of the clinical screening for signs of postnatal depression and anxiety
	PNC.B3a.DE.6–9	Mother–newborn cohabitation	Indicate whether the mother and newborn are residing together or separated
	PNC.B3a.DE.10–16	Newborn's current location	Specify where the newborn is (if not with mother/carer during PNC visit)
	PNC.B3a.DE.17–22	Breastfeeding status	Specify the details of current feeding practice
	PNC.B3a.DE.23–34	Breastfeeding / breast issues	Indicate whether the woman reports any problems breastfeeding and/or any discomfort/symptoms of the breasts
	PNC.B3a.DE.35–59	Specific complaint for the last 24 hours	Indicate whether the woman reports any of the following complaints in the past 24 hours
	PNC.B3a.DE.60–84	Vital signs from the past 24 hours	Specify the details of the mother's vital signs from the past 24 hours
	PNC.B3a.DE.85–87	Bowel movements after childbirth	Indicate whether the woman had a bowel movement after childbirth
	PNC.B3a.DE.88–90	Urine voided after childbirth	Indicate whether the woman voided her bladder after childbirth

Activity ID. Activity name	Data element ID	Data element label	Description and definition
	PNC.B3a.DE.91–93	Urinary incontinence or urinary leakage	Indicate whether the woman has experienced any symptoms of urinary incontinence or urinary leakage in the past 24 hours
	PNC.B3a.DE.94–96	Faecal incontinence	Indicate whether the woman has experienced any symptoms of faecal incontinence in the past 24 hours
	PNC.B3a.DE.97–126	Current medications	Select all of the medications the woman is currently taking
	PNC.B3a.DE.127–134	HIV catch-up testing after birth (including counselling)	Indicate whether the woman received HIV catch-up counselling and rapid test during today's contact and record the results
	PNC.B3a.DE.135–143	TB screening (including counselling)	Indicate whether the woman received counselling on TB screening during today's and record the results of the woman's TB screening
	PNC.B3a.DE.144–165	Postpartum contraception (counselling and method selected)	Indicate whether the woman received counselling on postpartum contraception and method selected
PNC.B4a. Conduct physical exam within 24 hours after birth (maternal)	PNC.B4a.DE.1–12	Woman's danger signs	Specify if the woman has any danger signs, which are continuously monitored in the first 24 hours
	PNC.B4a.DE.13–16	Height and weight	Specify the details of the woman's biometric assessment
	PNC.B4a.DE.17–44	Vital signs	Specify the details of the woman's vital signs
	PNC.B4a.DE.45–55	Respiratory exam result	Specify the details of the woman's respiratory condition based on respiratory exam
	PNC.B4a.DE.56–67	Cardiac exam result	Specify the details of the woman's cardiac condition based on cardiac exam
	PNC.B4a.DE.68–80	Breast exam result	Specify the details of the woman's breast examination
	PNC.B4a.DE.81–88	Abdominal exam result	Specify the details of the woman's abdominal examination
	PNC.B4a.DE.89–103	Uterus assessment	Specify the details of woman's uterine assessment
	PNC.B4a.DE.104–112	Pelvic exam results	Specify the details of woman's pelvic examination
	PNC.B4a.DE.113–139	Perineal examination	Specify the details of woman's perineal examination
	PNC.B4a.DE.140–156	Caesarean-section incision/wound examination	Specify the details of the woman's caesarean-section incision/wound examination
	PNC.B4a.DE.157–170	Presence of oedema	Indicate whether the woman presents with oedema
PNC.B5. OR PNC.D5. Emergency referral needed?	See PNC.C: Emergency referral		
PNC.B6a. OR PNC.D6a. Provide counselling, management and treatment (maternal)	PNC.B6a.DE.1–31	Suggested diagnosis(es)	Indicates potential diagnosis(es) and related treatment(s) for the woman
	PNC.B6a.DE.32	Counselling for local cooling for perineal pain relief	Whether counselling was provided for local cooling for perineal pain relief
	PNC.B6a.DE.33	Counselling for pharmacological relief for perineal pain	Whether counselling was provided for pharmacological relief for perineal pain

Activity ID. Activity name	Data element ID	Data element label	Description and definition
	PNC.B6a.DE.34	Counselling for pharmacological relief of pain due to uterine cramping	Whether counselling was provided for pharmacological relief of pain due to uterine cramping
	PNC.B6a.DE.35	Dietary advice and information to prevent constipation	Whether dietary advice and information to prevent constipation was provided
	PNC.B6a.DE.36	Preventive anthelmintic treatment	Whether preventive anthelmintic treatment was given
	PNC.B6a.DE.37–38	Preventive schistosomiasis treatment and counselling	Whether preventive schistosomiasis treatment and counselling was given/provided
	PNC.B6a.DE.39	Counselling on tobacco cessation	Whether counselling was provided on tobacco cessation
	PNC.B6a.DE.40	Counselling on exposure to second-hand smoke	Whether counselling was provided on exposure to second-hand smoke
	PNC.B6a.DE.41	Counselling on alcohol/substance use	Whether counselling was provided on alcohol/substance use
	PNC.B6a.DE.42	Counselling on condom use	Whether counselling was provided on condom use
	PNC.B6a.DE.43	Counselling on high risk for HIV infection and prescription of pre-exposure prophylaxis (PrEP)	Whether counselling was provided on high risk for HIV infection
	PNC.B6a.DE.44	Counselling on physical activity and sedentary behaviour	Whether counselling was provided on physical activity and sedentary behaviour
	PNC.B6a.DE.45	Counselling on postpartum oral iron and folate supplementation	Whether counselling was provided on postpartum oral iron and folate supplementation
	PNC.B6a.DE.46	Counselling for treatment of breast engorgement	Whether counselling was provided for non-pharmacological treatment of breast engorgement
	PNC.B6a.DE.47	Counselling for prevention of postpartum mastitis	Whether counselling was provided for non-pharmacological interventions to prevent postpartum mastitis
	PNC.B6a.DE.48	Counselling on breastfeeding to a mother who is apart from the newborn	Whether counselling was provided to a client who is apart from the newborn
	PNC.B6a.DE.49	Counselling on psychosocial interventions to prevent postpartum depression and anxiety	Whether counselling was provided on psychosocial interventions to prevent postpartum depression and anxiety

Activity ID. Activity name	Data element ID	Data element label	Description and definition
	PNC.B6a.DE.50	Counselling after neonatal death	Whether counselling was provided to a client whose newborn died
	PNC.B6a.DE.51	Counselling conducted on length of stay	Counselling provided on length of stay in health facility
	PNC.B6a.DE.52	Assessment of criteria prior to discharge from health facility conducted	Assessment of criteria prior to discharge from health facility conducted
	PNC.B6a.DE.53	Counselling on preparation for discharge conducted	Whether counselling was provided on preparation for discharge
	PNC.B6a.DE.54	Counselling conducted on seeking care when maternal danger signs occur	Whether counselling was provided on when and how to seek care when maternal danger signs are present
	PNC.B6a.DE.55	Counselling on PNC contact schedule conducted	Whether counselling was provided on PNC contact schedule conducted
PNC.B7. OR PNC.D7. Non-emergency referral	PNC.B7.DE.1	Non-emergency referral (maternal)	Referral for non-emergency care (maternal)
	PNC.B7.DE.2	Non-emergency referral (neonatal/infant)	Referral for non-emergency care (neonate/infant)
	PNC.B7.DE.3–6	Reason for non-emergency referral	Reason why the client is being referred
	PNC.B7.DE.7–9	Date of scheduled referral appointment	When the referral is scheduled
	PNC.B7.DE.10–12	Provider who made referral	The name of the provider who made the referral
	PNC.B7.DE.13–14	Referral notes	Any additional relevant details of clinical significance for the referral facility to provide quality care
Business process PNC.C: Emergency referral			
PNC.C1. Emergency referral	PNC.C1.DE.3–15	Reason for referral (maternal)	Reason why the patient (woman) is being referred
	PNC.C1.DE.16–25	Reason for referral (neonate/infant)	Reason why the patient (newborn/infant) is being referred
	PNC.C1.DE.26–28	Provider who made referral	The name of the provider who made the referral
	PNC.C1.DE.29–30	Referral notes	Any additional relevant details of clinical significance for the referral facility to provide quality care.
Business process PNC.D: Follow-up PNC contact			
PNC.D1. Review information from past PNC contact(s)	N/A—No data are recorded during this activity		
PNC.D2. Update profile and health history information	See PNC.B1. Record profile and health history		

Activity ID. Activity name	Data element ID	Data element label	Description and definition
PNC.D3a. Check symptoms and well-being during follow-up contact (maternal)	PNC.D3a.DE.1–5	Screening for postnatal depression and anxiety	Specify the results of the clinical screening for signs of postnatal depression and anxiety
	PNC.D3a.DE.6–9	Mother–newborn/infant pair attendance	Indicate whether the woman attends the postnatal care contact with the newborn/infant
	PNC.D3a.DE.10–13	Newborn's/infant's current location	Specify where the newborn is (if not with mother/carer during PNC visit)
	PNC.D3a.DE.14–19	Breastfeeding status	Specify the details of current feeding practice
	PNC.D3a.DE.20–29	Breastfeeding/breast issues	Indicate whether the woman reports any problems breastfeeding and/or discomfort/symptoms of the breasts
	PNC.D3a.DE.30–51	Specific complaint	Indicate whether the woman reports any of the following complaints between last contact and today
	PNC.D3a.DE.52–55	Regular bowel movements	Indicate whether the woman reports regular bowel movements
	PNC.D3a.DE.56–59	Regular urination	Indicate whether the woman reports regular urination
	PNC.D3a.DE.60–63	Urinary incontinence or urinary leakage	Indicate whether the woman has experienced any symptoms of urinary incontinence or urinary leakage
	PNC.D3a.DE.64–67	Faecal incontinence	Indicate whether the woman has experienced any symptoms of faecal incontinence
	PNC.D3a.DE.68–97	Current medications	Select all of the medications the woman is currently taking
PNC.D4a. Conduct physical exam during follow-up contact (maternal)	PNC.D4a.DE.1–4	Height and weight	Specify the details of the woman's biometric assessment
	PNC.D4a.DE.5–26	Vital signs	Specify the details of the woman's vital signs
	PNC.D4a.DE.27–37	Respiratory exam result	Specify the details of the woman's respiratory condition based on respiratory exam
	PNC.D4a.DE.38–49	Cardiac exam result	Specify the details of the woman's cardiac condition based on cardiac exam
	PNC.D4a.DE.50–62	Breast exam result	Specify the details of the woman's breast examination
	PNC.D4a.DE.63–70	Abdominal exam result	Specify the details of the woman's abdominal examination
	PNC.D4a.DE.71–85	Uterus assessment	Specify the details of woman's uterus assessment
	PNC.D4a.DE.86–94	Pelvic exam result	Specify the details of the woman's pelvic examination
	PNC.D4a.DE.95–121	Perineal examination	Specify the details of the woman's perineal examination
	PNC.D4a.DE.122–138	Caesarean-section incision/wound examination	Specify the details of the woman's caesarean-section incision/wound examination
	PNC.D4a.DE.139–152	Presence of oedema	Indicate whether the woman presents with oedema
PNC.D5. Emergency referral needed?	See PNC.C: Emergency referral		
PNC.D6a. Provide counselling, management and treatment (maternal)	See PNC.B6a: Provide counselling, management and treatment (maternal)		
PNC.D7. Non-emergency referral	See PNC.B7: Non-emergency referral		

Table 8. Workflow core data elements for identified PNC business processes – newborn/infant care

Activity ID. Activity name	Data element ID	Data element name	Description and definition
Business process PNC.B: PNC contact within first 24 hours after birth			
PNC.B3b. Check symptoms and well-being within 24 hours after birth (neonatal)	PNC.B3b.DE.1–5	Is the newborn with the mother?	Indicate whether the mother attends the postnatal care contact with the newborn. Specify where the mother is if she does not attend the PNC visit.
	PNC.B3b.DE.6–23	Complications/danger signs in the last 24 hours	Indicate whether the newborn has experienced any of the following complications/ danger signs in the last 24 hours
	PNC.B3b.DE.24–27	Vitamin K administered at birth	Indicate whether vitamin K injection was administered at birth
	PNC.B3b.DE.28–34	Vaccines administered at birth	Indicate whether the following doses of vaccine were administered at birth
	PNC.B3b.DE.35–40	Newborn's blood type ABO	Specify the newborn's blood type ABO
	PNC.B3b.DE.41–44	Newborn's blood type Rh	Specify the newborn's blood type Rh
	PNC.B3b.DE.45–50	Feeding status	Specify the details of current feeding status
	PNC.B3b.DE.51–53	Feeding frequency	Specify the details of current feeding frequency
	PNC.B3b.DE.54–59	Feeding mode	Specify the details of feeding mode
	PNC.B3b.DE.60–66	Has the newborn received any pre-lacteal feeds?	Indicate whether any of the following have been given to the newborn before breastfeeding initiation
	PNC.B3b.DE.67–73	Breastfeeding issues	Indicate any problems breastfeeding due to the following issues
	PNC.B3b.DE.74–76	Does the mother/family member feel motivated to take care of the newborn?	Indicate whether the mother/family member reports motivation to care for the newborn
	PNC.B3b.DE.77–81	Perception of newborn's quality of sleep	Specify the reported perception of the newborn's sleep quality
	PNC.B3b.DE.82–86	Where is the newborn sleeping?	Specify the location where the newborn usually sleeps
	PNC.B3b.DE.87–91	Sleeping position	Specify the position in which the newborn usually sleeps
	PNC.B3b.DE.92–95	Has the newborn been bathed?	Indicate whether the newborn has been bathed within the first 24 hours after birth
	PNC.B3b.DE.96–100	How was the umbilical cord stump cared for?	Indicate what care is provided to the umbilical cord stump
	PNC.B3b.DE.101–115	Vitals signs from the past 24 hours	Specify the details of the newborn vital signs from the past 24 hours
	PNC.B3b.DE.116–118	Bowel movements in the last 24 hours	Indicate whether the newborn has had a bowel movement in the past 24 hours
	PNC.B3b.DE.119–122	Urine voided in their last 24 hours	Indicate whether the newborn has voided their bladder in the past 24 hours

Activity ID. Activity name	Data element ID	Data element name	Description and definition
PNC.B4b. Conduct physical exam within 24 hours after birth (neonatal)	PNC.B4b.DE.1–18	Newborn's danger signs	Specify if the newborn has any danger signs, which are continuously monitored in the first 24 hours
	PNC.B4b.DE.19–24	Appearance	Specify the general appearance of the newborn
	PNC.B4b.DE.25–30	Skin	Specify the details of the examination of the newborn's skin
	PNC.B4b.DE.31–38	Jaundice location	Specify the location of jaundice
	PNC.B4b.DE.39–44	Spontaneous movements	Specify the characteristics of the newborn's spontaneous movements
	PNC.B4b.DE.45–47	Tone	Specify the characteristics of the newborn's muscle tone
	PNC.B4b.DE.48–51	Warmth	Specify whether the newborn is being kept at an appropriate temperature
	PNC.B4b.DE.52–59	Malformation	Specify if the newborn has any malformation
	PNC.B4b.DE.60–71	Vital signs and measurements	Specify the details of the newborn's vital signs and other clinical measurements
	PNC.B4b.DE.72–79	Head	Specify the details of the examination of the newborn's head
	PNC.B4b.DE.80–83	Fontanelles	Specify the details of the examination of the newborn's fontanelles
	PNC.B4b.DE.84–87	Face	Specify the details of the examination of the newborn's face
	PNC.B4b.DE.88–96	Eyes	Specify the details of the examination of the newborn's eyes
	PNC.B4b.DE.97–101	Mouth and palate	Specify the details of the examination of the newborn's mouth and palate
	PNC.B4b.DE.102–104	Neck and clavicles	Specify the details of the examination of the newborn's neck and clavicles
	PNC.B4b.DE.105–107	Heart	Specify the details of the examination of the newborn's heart
	PNC.B4b.DE.108–110	Lungs	Specify the details of the examination of the newborn's lungs
	PNC.B4b.DE.111–120	Abdomen	Specify the details of the examination of the newborn's abdomen
	PNC.B4b.DE.121–127	Umbilical cord	Specify the details of the examination of the newborn's umbilical cord
	PNC.B4b.DE.128–141	Genitalia	Specify the details of the examination of the newborn's genitalia
	PNC.B4b.DE.142–146	Anus	Specify the details of the examination of the newborn's anal area
	PNC.B4b.DE.147–151	Limbs	Specify the details of the examination of the newborn's limbs
	PNC.B4b.DE.152–160	Back and spine	Specify the details of the examination of the newborn's back and spine
	PNC.B4b.DE.161–169	Breastfeeding / feeding evaluation	Indicate whether an assessment of the feeding practice of the newborn was conducted. If done, specify positioning and suckling.
	PNC.B4b.DE.170–173	Screening for hearing impairment (otoacoustic emissions [OAE])	Specify the details of the newborn's screening for hearing impairment with OAE
	PNC.B4b.DE.174–177	Screening for neonatal hyperbilirubinaemia	Specify the details of the newborn's screening for neonatal hyperbilirubinaemia with transcutaneous bilirubinometer (TcB)

Activity ID. Activity name	Data element ID	Data element name	Description and definition
PNC.B6b. OR D6b. Provide counselling, management and treatment (newborn/infant)	PNC.B6b.DE.1–39	Suggested diagnoses	Indicates potential diagnosis(es) and related treatment(s) for the newborn/infant
	PNC.B6b.DE.40	Counselling and screening for abnormalities of the eye	Whether counselling on and screening for abnormalities of the eye was provided to the client
	PNC.B6b.DE.41	Counselling and screening for hearing impairment	Whether counselling on and screening for hearing impairment was provided to the client
	PNC.B6b.DE.42	Counselling and screening for neonatal hyperbilirubinaemia	Whether counselling on and screening for neonatal hyperbilirubinaemia was provided to the client
	PNC.B6b.DE.43	Counselling on timing of first bath	Whether counselling was provided on delaying first bath for at least 24 hours to prevent hypothermia and its sequelae
	PNC.B6b.DE.44	Counselling on umbilical cord care	Whether counselling was provided on dry, clean umbilical cord care
	PNC.B6b.DE.45	Counselling on sleeping position	Whether counselling was provided on sleeping position for the prevention of sudden infant death syndrome
	PNC.B6b.DE.46	Counselling for newborn immunization	Whether counselling was provided for newborn immunization
	PNC.B6b.DE.47	Counselling on vitamin A supplementation	Whether counselling was provided on supplementation with vitamin A
	PNC.B6b.DE.48	Counselling on how to breastfeed	Whether counselling was provided on breastfeeding
	PNC.B6b.DE.50	Counselling on breast/breastfeeding issues	Whether counselling was provided on breast/breastfeeding issues
	PNC.B6b.DE.53	Counselling on responsive care and early learning activities	Whether counselling was provided on responsive care and early learning
	PNC.B6b.DE.54	Counselling on gentle whole-body massage	Whether counselling was provided on gentle whole-body massage
	PNC.B6b.DE.56	Assessment of criteria prior to discharge from health facility conducted	Assessment of criteria prior to discharge from health facility conducted
	PNC.B6b.DE.57	Counselling on preparation for discharge conducted	Whether counselling was provided on preparation for discharge
	PNC.B6b.DE.58	Counselling on seeking care when newborn danger signs occur	Whether counselling was provided on when and how to seek care when danger signs are present

Activity ID. Activity name	Data element ID	Data element name	Description and definition
Business process PNC.D: PNC follow-up contact			
PNC.D3b. Check symptoms and well-being during follow up contact (neonatal/infant)	PNC.D3b.DE.1–7	Is the newborn/infant with the mother?	Indicate whether the mother attends the PNC contact with the newborn/infant
	PNC.D3b.DE.8–26	Complications/danger signs between last contact and today	Indicate whether the newborn/infant has experienced any of the following complications/danger signs between last contact and today
	PNC.D3b.DE.27–33	Vaccines administered since last contact	Indicate whether the following doses of vaccine have been administered since last contact
	PNC.D3b.DE.34–39	Feeding status	Specify the details of current feeding status
	PNC.D3b.DE.40–42	Feeding frequency	Specify the details of current feeding frequency
	PNC.D3b.DE.43–48	Feeding mode	Specify the details of feeding mode
	PNC.D3b.DE.49–55	Breastfeeding issues	Indicate any problems breastfeeding due to the following issues
	PNC.D3b.DE.56–58	Does the mother/family member feel motivated to take care of the newborn/infant?	Indicate whether the mother/family member reports motivation to care for the newborn/infant
	PNC.D3b.DE.59–63	Perception of newborn's/infant's quality of sleep	Specify the reported perception of the newborn's/infant's sleep quality
	PNC.D3b.DE.64–68	Where is the newborn/infant sleeping?	Specify the location where the newborn/infant usually sleeps
	PNC.D3b.DE.69–73	Sleeping position	Specify the position in which the newborn/infant usually sleeps
	PNC.D3b.DE.74–78	How was the umbilical cord stump cared for?	Indicate what care is provided to the umbilical cord stump
	PNC.D3b.DE.79–82	Bowel movements	Indicate whether the newborn/infant has regular bowel movements
	PNC.D3b.DE.83–86	Regular urination	Indicate whether the newborn/infant has voided their bladder

Activity ID. Activity name	Data element ID	Data element name	Description and definition
PNC.D4b. Conduct physical exam during follow up contact (neonatal/infant)	PNC.D4b.DE.1–6	Appearance	Specify the general appearance of the newborn/infant
	PNC.D4b.DE.7–12	Skin	Specify the details of the examination of the newborn's/infant's skin
	PNC.D4b.DE.13–21	Jaundice location	Specify the location of jaundice
	PNC.D4b.DE.22–27	Spontaneous movements	Specify the characteristics of the newborn's/infant's spontaneous movements
	PNC.D4b.DE.28–30	Tone	Specify the characteristics of the newborn/infant muscle tone
	PNC.D4b.DE.31–34	Warmth	Specify whether the newborn/infant is being kept at an appropriate temperature
	PNC.D4b.DE.35–42	Malformation	Specify if the newborn/infant has any malformation
	PNC.D4b.DE.43–54	Vital signs and measurements	Specify the details of the newborn's/infant's vital signs and other clinical measurements
	PNC.D4b.DE.55–62	Head	Specify the details of the examination of the newborn's/infant's head
	PNC.D4b.DE.63–66	Fontanelles	Specify the details of the examination of the newborn's/infant's fontanelles
	PNC.D4b.DE.67–70	Face	Specify the details of the examination of the newborn's/infant's face
	PNC.D4b.DE.71–79	Eyes	Specify the details of the examination of the newborn's/infant's eyes
	PNC.D4b.DE.80–84	Mouth and palate	Specify the details of the examination of the newborn's/infant's mouth and palate
	PNC.D4b.DE.85–87	Neck and clavicles	Specify the details of the examination of the newborn's/infant's neck and clavicles
	PNC.D4b.DE.88–90	Heart	Specify the details of the examination of the newborn's/infant's heart
	PNC.D4b.DE.91–93	Lungs	Specify the details of the examination of the newborn's/infant's lungs
	PNC.D4b.DE.94–103	Abdomen	Specify the details of the examination of the newborn's/infant's abdomen
	PNC.D4b.DE.104–109	Umbilical cord	Specify the details of the examination of the newborn's umbilical cord
	PNC.D4b.DE.110–119	Genitalia	Specify the details of the examination of the newborn's/infant's genitalia
	PNC.D4b.DE.120–124	Anus	Specify the details of the examination of the newborn's/infant's anal area
	PNC.D4b.DE.125–129	Limbs	Specify the details of the examination of the newborn's/infant's limbs
	PNC.D4b.DE.130–138	Back and spine	Specify the details of the examination of the newborn's/infant's back and spine
	PNC.D4b.DE.139–147	Breastfeeding/feeding evaluation	Indicate whether an assessment of the feeding practice of the newborn/infant was conducted. If done, specify positioning and suckling
	PNC.D4b.DE.148–151	Screening for hearing impairment (otoacoustic emissions [OAE])	Specify the details of the newborn's/infant's screening for hearing impairment with OAE

5.2 List of calculated data elements

Section 5.1 outlines the core data elements that should be included within digital systems to facilitate the decision-support logic or indicators. There are additional derived data elements that are based on calculations from core data elements (**Table 9**).

Table 9. Calculated data elements

Calculated data element label	Core data elements used for calculation (i.e. the “variables”)	Calculation
Body Mass Index (BMI)	Body weight Height	$\text{“Body weight (kg)”} / [\text{“Height (cm)”} / 100] ^2$
Postpartum (Woman)	Date of childbirth Visit date	IF “Visit date” – “Date of childbirth” ≤ 6 weeks, THEN “Postpartum” = TRUE

5.3 Additional considerations for adapting the data dictionary

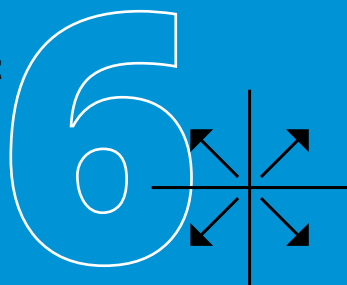
Some settings may require the inclusion of additional data elements into the full data set or changes to response options based on contextual differences. Additionally, the transition from paper-based forms to digital systems may require some reflection on whether data elements currently on the paper forms should be incorporated into the digital system. **Table 10** is an initial list of considerations anticipated for each implementation to review and customize based on the national guidelines and local context. The annex provides additional guidance for adding data elements to or amending existing data elements in the data dictionary.

Table 10. Characteristics for local customization and configuration

Data element ID	Point of customization	Description
Population characteristics		
PNC.Config.1	Population prevalence of gestational anaemia	The proportion of pregnant women in the population with anaemia (haemoglobin level < 11 g/dl)
PNC.Config.2	Population prevalence of tuberculosis (TB) per 100 000 persons	The TB prevalence in the general population in number of cases per 100 000 persons
PNC.Config.3	Population prevalence of tuberculosis (TB) percentage	The TB prevalence in the general population in number of cases per 100 (percentage)
PNC.Config.4	Population prevalence of soil-transmitted helminth infection	The percentage of individuals in the general population infected with at least one species of soil-transmitted helminth
PNC.Config.5	Population incidence of HIV in the absence of pre-exposure prophylaxis (PrEP)	HIV incidence: number of cases per 100 person-years
PNC.Config.6	Population prevalence of HIV in key populations (men who have sex with men, people in prison or other closed settings, people who inject drugs, sex workers and transgender people)	The proportion of individuals in each of the key populations who are HIV positive
PNC.Config.7	Population prevalence of HIV in pregnant women	The proportion of pregnant women in the population who are HIV positive

PNC.Config.8	Malaria-endemic setting	Whether the setting is a malaria-endemic setting
PNC.Config.9	Schistosomiasis-endemic setting	Whether the setting is a schistosomiasis-endemic setting
PNC.Config.10	Population prevalence of <i>Schistosoma</i> spp. infection	The prevalence of <i>Schistosoma</i> spp. in endemic communities
PNC.Config.11	Infant mortality rate	The occurrence of infant mortality (children under 1 year old) per 1000 live births
PNC.Config.12	Population prevalence of maternal vitamin A deficiency	The proportion of pregnant women with serum retinol concentrations < 0.70 µmol/L
Site-specific characteristics		
PNC.Config.13	Prevalence of HIV in pregnant women in the catchment area	The proportion of pregnant women in the health facility's catchment area who are HIV positive, based on the number of pregnant women attending antenatal clinics at the health facility

Component



Decision-support logic

The decision-support logic component of the DAK provides the decision logics and algorithms, as well as the scheduling of services, in accordance with WHO guidelines. In this DAK, the decision logics and algorithms deconstruct the recommendations within the WHO PNC guideline (19) into a format that clearly labels the inputs and outputs that would be operationalized in a digital decision-support system.

6.1 Decision-support logic overview

Tables 11 and 12 provide an overview of the decision-support tables and algorithms for the different PNC module business processes. The structure of the decision-support tables is based on an adaptation of the Decision Model and Notation (DMN), an industry standard for modelling and executing decision logics (53). These decision-support tables detail the business rules, data inputs and outputs to support PNC module business processes.

Table 11. Overview of decision-support tables for PNC – woman

Activity ID. Activity name	Decision-support table ID	Decision-support table description	Reference/source(s)
PNC.A2.DT.1 Reception and assessment of danger signs	PNC.A2.DT.1	Assess the general condition of the woman immediately on arrival or during the physical examination.	Pregnancy, childbirth, postpartum and newborn care guide (2015) (34)
PNC.A4.DT.2 Collect client details – conduct HEADSS assessment	PNC.A4.DT.2	Conduct a Home, Education/Employment, Eating, Activity, Drugs, Sexuality, Safety, Suicide/Depression (HEADSS) assessment.	<i>Adolescent job aid: a handy desk reference tool for primary level health workers</i> (2010) (51)
PNC.B3a.DT.3 or PNC.D3a.DT.3 Check symptoms and well-being (maternal)	PNC.B3a.DT.3	Assess woman's status: – mental health screening for depression/anxiety – breastfeeding status – micturition and urinary incontinence – bowel function and faecal incontinence – current medications.	WHO PNC recommendations (2022) (19): A1
PNC.B3a.DT.4 Check symptoms and well-being within 24 hours after birth (maternal) – HIV catch-up testing and TB screening	PNC.B3a.DT.4	Proceed with testing depending on woman's risk: – HIV catch-up testing – TB screening.	WHO PNC recommendations (2022) (19): A.2a, A.2b, A.3a–3c

Activity ID. Activity name	Decision-support table ID	Decision-support table description	Reference/source(s)
PNC.B3a.DT.5 Check symptoms and well-being within 24 hours after birth (maternal) – contraception	PNC.B3a.DT.5	Provision of comprehensive family planning information and services.	WHO PNC recommendations (2022) (19): A24 Pregnancy, childbirth, postpartum and newborn care guide (2015) (34): C16 <i>Medical eligibility criteria for contraceptive use</i> (2015) (46)
PNC.B4a.DT.6 or PNC.D4a. Conduct a physical exam (maternal)	PNC.B4a.DT.6	Conduct physical examination. Depending on the results of the physical exam, additional examination and interventions may be needed.	WHO PNC recommendations (2022) (19): A1 Managing complications guide (2017) (32)
PNC.B6a.DT.7 or PNC.D6a. Provide counselling, management and treatment for woman – suggested diagnosis and treatment	PNC.B6a.DT.7	Conduct management, counselling and/or referral as needed, based on the diagnosis presented by the woman.	WHO PNC recommendations (2022) (19): A.2a, A.2b, A3a–3c Pregnancy, childbirth, postpartum and newborn care guide (2015) (34)
PNC.B6a.DT.8 or PNC.D6a. Provide counselling, management and treatment for woman – interventions for physiological signs and symptoms	PNC.B6a.DT.8	Provide interventions for specific signs and symptoms: – perineal pain relief – uterine cramping/involution – breast engorgement.	WHO PNC recommendations (2022) (19): A.4–6, A.8
PNC.B6a.DT.9 or PNC.D6a. Provide counselling, management and treatment for woman – risk reduction	PNC.B6a.DT.9	Provide preventive measures for the woman for: – anthelmintic/schistosomiasis treatment – physical activity and sedentary behaviour – postpartum mastitis – postpartum constipation – postpartum depression and anxiety.	WHO PNC recommendations (2022) (19): A.10, A.15–17
PNC.B6a.DT.10 or PNC.D6a. Provide counselling, management and treatment for woman – general counselling	PNC.B6a.DT.10	Provide counselling on postpartum period. Depending on the woman's reported health-related behaviours or risk conditions identified during the contact, additional counselling may or may not be required.	WHO PNC recommendations (2022) (19): A.20, C.45, C.46 WHO ANC recommendations (2016) (45): B.1.5–B.1.7, C.7 Pregnancy, childbirth, postpartum and newborn care guide (2015) (34) <i>Pregnancy management with Zika</i> (2016) (54)
PNC.C1.DT.11 Emergency referral	PNC.C1.DT.11	Physical signs and symptoms and exam results requiring emergency referral.	Pregnancy, childbirth, postpartum and newborn care guide (2015) (34) WHO PNC recommendations (2022) (19): A.1 Managing complications guide (2017) (32) WHO pre-eclampsia and eclampsia recommendations (2011) (55): 4, 6, 7

HEADSS: home, education/employment, activities, drugs, sex and relationships, self-harm and depression, safety and abuse.

Table 12. Overview of decision-support tables for PNC – newborn/infant

Activity ID. Activity name	Decision-support table ID	Decision-support table description	Reference/source(s)
PNC.A2.DT.12 Reception and assessment of danger signs	PNC.A2.DT.12	Assess the general condition of the newborn/infant immediately on arrival or during the physical examination.	Pregnancy, childbirth, postpartum and newborn care guide (2015) (34) WHO PNC recommendations (2022) (19): B.25
PNC.B3b.DT.13 or PNC.D3b.DT.13 Check symptoms and well-being (neonatal/ infant)	PNC.B3b.DT.13	Assess the newborn's/infant's status: – vaccines – feeding status, frequency and mode – quality of sleep and sleeping position – delayed bathing (within 24 hours after birth only) – umbilical cord stump care – urinary and bowel functions.	WHO PNC recommendations (2022) (19): B.25, C.46 <i>Early essential newborn care: clinical practice pocket guide</i> (2022) (31) Pregnancy, childbirth, postpartum and newborn care guide (2015) (34)
PNC.B4b.DT.14 or PNC.D4b.DT.14 Conduct physical exam of the newborn/infant	PNC.B4b.DT.14	Conduct physical examination. Depending on the results of some physical exams, additional examinations or repeat measurements are required.	Managing complications guide (2017) (32) Pregnancy, childbirth, postpartum and newborn care guide (2015) (34)
PNC.B4b.DT.15 Conduct physical exam of the newborn within 24 hours after birth – neonatal screening	PNC.B4b.DT.15	Conduct the following screenings: – screening for abnormalities of the eye – universal screening for hearing impairment – universal screening for neonatal hyperbilirubinaemia.	WHO PNC recommendations (2022) (19): B.26–29
PNC.B6b.DT.16 or PNC.D6b.DT.16 Provide counselling, management and treatment for newborn/ infant – suggested diagnosis and treatment	PNC.B6b.DT.16	Conduct management, counselling or referral as needed, based on the diagnosis presented by the newborn/infant.	Pregnancy, childbirth, postpartum and newborn care guide (2015) (34) WHO PNC recommendations (2022) (19): B.4–5 <i>Early essential newborn care: clinical practice pocket guide</i> (2022) (31) Pocket book of hospital care for children (2013) (35)
PNC.B6b.DT.17 or PNC.D6b.DT.17 Provide counselling, management and treatment for newborn/ infant – risk reduction and general counselling	PNC.B6b.DT.17	Provide counselling on preventive measures such as: – timing of first bath (within 24 hours after birth only) – hygiene of umbilical cord – sleeping position – newborn/infant immunization.	WHO PNC recommendations (2022) (19): B.30, B.33 <i>Early essential newborn care: clinical practice pocket guide</i> (2022) (31)
PNC.B6b.DT.18 or PNC.D6b.DT.18 Provide counselling, management and treatment for newborn/infant – nutritional interventions	PNC.B6b.DT.18	Provide counselling on nutritional interventions for the newborn/infant such as: – breastfeeding – vitamin A supplementation.	WHO PNC recommendations (2022) (19): B.35b, B.42 Pregnancy, childbirth, postpartum and newborn care guide (2015) (34)
PNC.C1.DT.19 Emergency referral	PNC.C1.DT.19	Physical signs and symptoms and exam results requiring emergency referral.	Pregnancy, childbirth, postpartum and newborn care guide (2015) (34) <i>Early essential newborn care: clinical practice pocket guide</i> (2022) (31) Pocket book of hospital care for children (2013) (35)

6.2 Decision-support tables

Each of the decision logics listed in the overview table is elaborated in the Implementation tools: [Decision-support logic](#). These decision-support tables comprise the components described in [Table 13](#).

Note that the decision-support logic here is translated directly from the WHO guidelines and guidance documents and has been reviewed by the panel of experts who created those guidelines documents. We do not anticipate the decision-support logic to change much as the logic has been created and reviewed by clinical experts. However, some level of adaptation may be needed depending on changes to the workflow and/or changes to the data dictionary.

Any changes to the decision-support logic should be considered carefully, as an embedded decision-support system can greatly affect the quality of care at the point of care. As helpful as decisions-support logic can be to the health worker, incorrect decision-support logic can also be detrimental. Thus, any new decision-support logic should be carefully reviewed and agreed upon by in-country clinical experts. For all the decision-support tables that are available for the PNC DAK, please refer to the linked spreadsheet Implementation tools: [Decision-support logic](#).

An example of a decision-support logic table for physical examination of the woman is given in [Table 14](#).

Table 13. Components of the decision-support tables

Decision ID	The name of the “decision” describing what algorithm or logic is represented (e.g. physical examination of the woman and related follow-up actions, if needed). The Decision ID should correspond to the number in Tables 11 and 12 .				
Business rule	The description of the decision that needs to be made based on IF/THEN statements with the appropriate data element label for the variables. The rule should demonstrate the relationship between the input variables and the expected outputs and actions within the decision logic.				
Trigger	The event that would indicate when this decision-support logic should appear within the workflow, such as the activity that would trigger this decision to be made.				
Input(s)	Output type	Action	Guidance	Annotations	
These inputs are the variables that need to be considered to determine the consequent actions or outputs.	If there are multiple input entries in the same row, these different inputs are considered as “AND” conditions that need to be in place at the same time.	The resulting action or decision based on the combination of input entries. This is the statement that immediately follows the “THEN”. Examples of outputs include a diagnosis, alerts/prompts for referral, counselling.	Concrete measures to be taken based on the output (e.g. refer, provide counselling, conduct test). In some cases, output and action may be the same.	Pop-up alert messages for the health worker. This cell should include the written content that would appear in the pop-up messages notifying the health worker of the appropriate next step(s).	This column should be used for any other notes, annotations or communication messages. This should include any additional information that does not fit into the other columns.
Inputs placed in different rows are considered as “OR” and can be considered independently of the inputs in other rows.					

Table 14. Example decision-support logic table for physical exam of the woman

Decision ID		PNC.B4a.DT.6				
Business rule		Conduct physical examination. Depending on the results of the physical exam, additional examinations or repeat measurements are required.				
Trigger		PNC.B4a. Conduct physical exam within 24 hours after birth (maternal) AND PNC.D4a. Conduct physical exam during follow-up contact (maternal)				
Hit policy indicator		Rule order				
Input(s)		Output	Action	Guidance	Annotations	References
“Systolic blood pressure” < 90 mmHg or “Systolic blood pressure” ≥ 140 mmHg		ActivityDefinition	Measure blood pressure (BP) again after 10–15 minutes’ rest	Given abnormal values (systolic blood pressure [SBP] ≥ 140 or < 90) at the first measurement, BP should be measured again.		<i>WHO recommendations on maternal and newborn care for a positive postnatal experience</i> (2022) (19): recommendation A1 Managing complications guide in pregnancy and childbirth (2017) (32)
“Diastolic blood pressure” < 60 mmHg or “Diastolic blood pressure” ≥ 90 mmHg		ActivityDefinition	Measure BP again after 10–15 minutes’ rest	Given abnormal values (diastolic blood pressure [DBP] ≥ 90 or < 60) at the first measurement, BP should be measured again.		<i>WHO recommendations on maternal and newborn care for a positive postnatal experience</i> (19): recommendation A1 Managing complications guide in pregnancy and childbirth (2017) (32)
“Body temperature” < 36 °C or “Body temperature” ≥ 38 °C		ActivityDefinition	Measure temperature again	Temperature < 36 °C or ≥ 38 °C. Measure temperature again. Consider measuring temperature orally.		<i>WHO recommendations on maternal and newborn care for a positive postnatal experience</i> (19): recommendation A1 Managing complications guide in pregnancy and childbirth (2017) (32)
Pulse rate < 60 bpm or Pulse rate > 100 bpm		ActivityDefinition	Measure pulse rate again after 10–15 minutes rest	Pulse rate < 60 or > 100 bpm: measure pulse rate again after 10–15 minutes.		<i>WHO recommendations on maternal and newborn care for a positive postnatal experience</i> (19): recommendation A1 Managing complications guide in pregnancy and childbirth (2017) (32)
“Respiratory exam result” ≠ “Normal”	“Respiratory exam result” ≠ “Not done”	ActivityDefinition	Conduct oximetry test	Woman has an abnormal respiratory exam result or respiratory distress. Conduct oximetry test.		<i>WHO recommendations on maternal and newborn care for a positive postnatal experience</i> (19): recommendation A1 Managing complications guide in pregnancy and childbirth (2017) (32)
“Breast examination result” = “Not done”	–	ActivityDefinition	Consider breast examination	It is recommended to assess breast condition during the postpartum period.		<i>WHO recommendations on maternal and newborn care for a positive postnatal experience</i> (19): recommendation A1 Managing complications guide in pregnancy and childbirth (2017) (32)

Decision ID		PNC.B4a.DT.6				
Business rule		Conduct physical examination. Depending on the results of the physical exam, additional examinations or repeat measurements are required.				
Trigger		PNC.B4a. Conduct physical exam within 24 hours after birth (maternal) AND PNC.D4a. Conduct physical exam during follow-up contact (maternal)				
Hit policy indicator		Rule order				
Input(s)		Output	Action	Guidance	Annotations	References
“Fundal tone” = “Flaccid”	–	ActivityDefinition	Proceed with further investigation	Flaccid fundal tone. Risk of bleeding or infection. – Check for history or signs of bleeding and infection. – Assess additional signs and symptoms. – Conduct further investigation and, if necessary, refer to a hospital or high-level assistance.		WHO recommendations on maternal and newborn care for a positive postnatal experience (19): recommendation A1 Managing complications guide in pregnancy and childbirth (2017) (32)
“Pelvic exam result” = “Not done”	–	ActivityDefinition	Consider pelvic examination	It is recommended to assess pelvic conditions during the postpartum period, especially to assess wounds, haematomas, lochia and secretions.		WHO recommendations on maternal and newborn care for a positive postnatal experience (19): recommendation A1 Managing complications guide in pregnancy and childbirth (2017) (32)
“Caesarean section wound inspection: findings” = “Not done”	–	ActivityDefinition	Consider caesarean section wound examination	It is recommended to assess caesarean section wound conditions during the postpartum period, especially to assess signs of bleeding and infection.		WHO recommendations on maternal and newborn care for a positive postnatal experience (19): recommendation A1 Managing complications guide in pregnancy and childbirth (2017) (32)

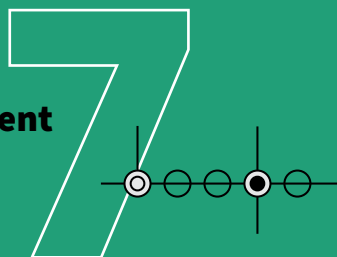
6.3 Scheduling logic overview

In addition to specific decision-support logic that needs to be detailed, there is also scheduling logic that can be used to facilitate the digital tracking of clients. For example, it will be important for the health worker to know when the client’s next visit is due based on the recommendations for follow-up. The follow-up schedule is based on the time since birth or the service that was provided. The overview of the follow-up schedules is provided in [Table 15](#) and the corresponding logic is elaborated in the Implementation tools: [Decision-support logic](#).

Table 15. Overview of scheduling logic

Scheduling logic ID	Scheduling logic description	Reference/source
PNC.S.01	PNC contact schedule	WHO PNC recommendations (2022) (19)
PNC.S.02	Newborn/infant immunization schedule	WHO PNC recommendations (2022) (19) <i>Digital adaptation kit for immunizations: operational requirements for implementing WHO recommendations in digital systems</i> (2024) (24)

Component



Indicators and performance metrics

This section details indicators and performance metrics that would be aggregated from core data elements in this digital system identified in [Component 5](#). The list in [Table 16](#) is a set of indicators that can be aggregated for decision-making and performance metrics of information in this digital system. These indicators are adapted from the minimum set of maternal, newborn/infant, child and adolescent health indicators and additional indicators for routine programme monitoring included in the *Analysis and use of health facility data: guidance for maternal, newborn, child and adolescent health programme managers* (28), *Indicators for assessing infant and young child feeding practices: definitions and measurement methods* (56), *Monitoring and evaluation for viral hepatitis B and C: recommended indicators and framework* (57), the global health observatory (58) and the Maternal, Newborn, Child, and Adolescent Health and Ageing Data Portal (59).

More details for calculating indicators are provided in the Implementation tool: [Indicators table](#).

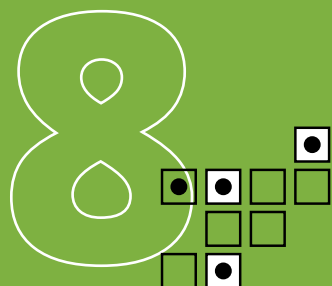
Table 16. PNC indicators

Indicator ID	Indicator name	Definition	Numerator description	Denominator description	Disaggregations	Reference(s)
<i>Postnatal care indicators</i>						
PNC.IND.1	PNC for women	Proportion of women with a PNC contact within a specified time period ^a after birth	Number of women with a PNC contact within the specified time period ^a after birth	Total number of women with any postnatal contact registered in the digital system	Timing of PNC contact in accordance with national policy Location (urban/rural) Mother's education level (none, don't know, primary, secondary, higher) Wealth quintile	Adapted from <i>Analysis and use of health facility data: guidance for maternal, newborn, child and adolescent health programme managers</i> (28)

PNC.IND.2	PNC for newborns/infants	Proportion of newborns/infants with a PNC contact within a specified time period ^a after birth	Number of newborns/infants with a PNC contact within the specified time period ^a after birth	Total number of newborns/infants with any postnatal contact registered in digital system	Timing of PNC contact in accordance with national policy Location (urban/rural) Mother's education level (none, don't know, primary, secondary, higher) Wealth quintile	<i>Analysis and use of health facility data: guidance for maternal, newborn, child and adolescent health programme managers (28)</i>
PNC.IND.3	Exclusive breastfeeding	Proportion of newborns/infants who were fed exclusively with breastmilk during the previous day	Number of newborns/infants who were fed exclusively with breastmilk during the previous day	Total number of newborns/infants with any postnatal contact registered in digital system	Newborn's/infant's age (< 28 days, 2–3 months) Newborn's/infant's sex Location (urban/rural) Mother's education level (none, don't know, primary, secondary, higher) Wealth quintile	<i>Indicators for assessing newborn/infant and young child feeding practices: definitions and measurement methods (56)</i>
PNC.IND.4	Postpartum family planning acceptor	Proportion of women who initiate or are currently using a modern contraceptive method	Number of women who initiate or are currently using a modern contraceptive method	Total number of women with any postnatal contact registered in the digital system	Mother's age (among adolescents [10–14, 15–19], by 5-year age groups [15–19, 20–24, 25–29, 30–34 years, etc.) Mother's education level Location (urban/rural)	<i>Analysis and use of health facility data: guidance for maternal, newborn, child and adolescent health programme managers (28)</i>
<i>Additional indicators</i>						
PNC.IND.5	Early initiation of breastfeeding	Proportion of newborns put to the breast within one hour of birth	Number of newborns born alive put to the breast within one hour of birth	Total number of newborns/infants with any postnatal contact registered in digital system	Newborn's/infant's sex Location (urban/rural)	<i>Analysis and use of health facility data: guidance for maternal, newborn, child and adolescent health programme managers (28)</i>
PNC.IND.6	Hepatitis B birth dose vaccination	Proportion of newborns/infants who have received the first dose of the hepatitis B vaccine within 24 hours after birth	Number of newborns/infants receiving birth dose of hepatitis B vaccine within 24 hours after birth	Total number of newborns/infants with any postnatal contact registered in digital system	Newborn's/infant's sex Location (urban/rural) Wealth quintile	<i>Monitoring and evaluation for viral hepatitis B and C: recommended indicators and framework (57)</i>

^a Timing of PNC may vary in accordance with national policy. The 2022 *WHO recommendations on maternal and newborn care for a positive postnatal experience* recommends a minimum of four PNC contacts (19). For births in health facilities, “healthy women and newborns should receive postnatal care in the facility for at least 24 hours after birth”. For births at home, the guideline recommends that “the first postnatal contact should be as early as possible within 24 hours of birth”. For healthy women and newborns/infants, the guideline recommends at least three additional PNC contacts: between 48 and 72 hours; between 7 and 14 days; and during week 6 after birth (19).

Component



High-level functional and non-functional requirements

This section provides an overview of illustrative functional and non-functional requirements that may be considered to kick-start the process of designing or adapting the digital tracking and decision-support system. Functional requirements describe the capabilities the system must have in order to meet the end-users' needs and achieve tasks within the business process. Non-functional requirements provide the general attributes and features of the digital system to ensure usability and overcome technical and physical constraints. Examples of non-functional requirements include ability to work offline, multiple language settings and password protection.

Table 17 highlights some key functional requirements for executing the business processes listed in **Component 4** of this document, and the complete set of functional requirements can be accessed in Implementation tool – *Functional and non-functional requirements*. **Table 18** provides non-functional requirements as general characteristics of the overall system. Please note that these lists are not exhaustive and should be modified according to context and user persona needs.

8.1 Functional requirements

Table 17. Functional requirements

Requirement ID	Activity ID and description	As a ...	I want ...	So that ...
Business process A: Registration				
PNC.FXNREQ.001	Assessment of danger signs	Health worker	To bypass the standard flow at any point if an urgent case is identified	The client(s) can receive immediate attention
PNC.FXNREQ.006	Search for client ^a name and record	Health worker or clerk	To see whether client is already in the system (using at least 2 identifiers)	I can check whether this is a new or existing client
PNC.FXNREQ.007	Search for client name and record	Health worker or clerk	The system to display sufficient data to identify the client	I can confirm that it is the correct client
PNC.FXNREQ.008	Search for client name and record	Health worker or clerk	The system to require me (a user) to search to see whether a client is already in the system prior to starting a new medical record entry	I can confirm that it is the correct client and update information as necessary
PNC.FXNREQ.009	Search for client name and record	Health worker or clerk	To use client identification system (e.g. QR code, barcode, fingerprint) to pull up client information	I can confirm that it is the correct client and update information as necessary

Requirement ID	Activity ID and description	As a ...	I want ...	So that ...
PNC.FXNREQ.010	Search for client name and record	Health worker or clerk	To provide sufficient data to rule out the possibility that this client is already in the system	I can avoid duplicates
PNC.FXNREQ.011	Create client record	Health worker or clerk	To be able to enter identification information	I can enter new client information or update an existing client record
PNC.FXNREQ.012	Create client record	Health worker or clerk	The system to indicate mandatory fields that must be filled out for registration to be valid	I can ensure all necessary information has been completed
PNC.FXNREQ.013	Create client record	Health worker or clerk	To generate encounter number for contact	I can initiate the required postnatal care services
PNC.FXNREQ.014	Create client record	Health worker or clerk	To generate or link to existing facility medical record number	I can check and confirm the information
PNC.FXNREQ.015	Create client record	Health worker or clerk	To edit fields on screen before information is committed	I can ensure information has been checked before submission
PNC.FXNREQ.016	Create client record	Health worker or clerk	To enter a temporary identification in emergency situations when full identity is unknown	I can proceed with registration
PNC.FXNREQ.017	Validate client details	Health worker or clerk	To display client information for validation (and be able to edit it)	I can ensure information has been checked before submission
PNC.FXNREQ.018	Validate client details	Health worker or clerk	To be able to update demographic information	The most current information on client can be recorded
PNC.FXNREQ.019	Validate client details	Health worker or clerk	To retain previous history of updated information	I can review past information
PNC.FXNREQ.020	Validate client details	Health worker or clerk	To be able to attach an identifier (e.g. QR code, barcode, fingerprint, photo) to the client's record based on national standards consent	I have additional ways of identifying the client
PNC.FXNREQ.021	Validate client details	Health worker or clerk	To be able to confirm client identity	I can be sure it is the right person
PNC.FXNREQ.022	Validate client details	Health worker or clerk	If this is a returning contact, to add the information to their previous contact	I can link the information across different contacts
PNC.FXNREQ.023	Check in client	Health worker or clerk	To record a time-and-date-stamped new contact (encounter)	I can confirm when the client came
Business process B: PNC contact within 24 hours after birth				
PNC.FXNREQ.024	Registration	Health worker or clerk	<i>See functional requirements above for registration (Business process A)</i>	N/A
PNC.FXNREQ.025	Record profile and health history	Health worker	The system to provide a standardized form for the entry of PNC profiles	I can ensure the woman's and/or newborn's/ infant's health information is recorded and up to date
PNC.FXNREQ.026	Record profile and health history	Health worker	To provide identification information for woman and/or newborn/infant based on data entered	I can update any demographic information

Requirement ID	Activity ID and description	As a ...	I want ...	So that ...
PNC.FXNREQ.027	Record profile and health history	Health worker	To print a copy of the pregnancy/birth information that replicates the paper postnatal register	I can more easily fulfil reporting requirements
PNC.FXNREQ.030	PNC assessment	Health worker	The system to provide a standardized form for the entry of clinical data	I can ensure the woman's and/or newborn's/ infant's health information is recorded and kept up to date
PNC.FXNREQ.031	PNC assessment	Health worker	The system to provide real-time range checks and integrity checks on data	I can ensure entry of accurate information and reduce errors
PNC.FXNREQ.032	PNC assessment	Health worker	To review past medical history previously entered	I can provide appropriate services according to the latest clinical protocols
PNC.FXNREQ.033	PNC assessment	Health worker	A list of tests that I (the health worker) can order	I can provide appropriate services according to the latest clinical protocols
PNC.FXNREQ.034	PNC assessment	Health worker	To print a requisition for lab tests, which includes required information for performing the test(s)	I can provide the woman and/or newborn/infant with list of tests required
PNC.FXNREQ.035	PNC assessment	Health worker	To provide decision support as appropriate for users based on data entered	I can provide appropriate services according to the latest clinical protocols
PNC.FXNREQ.036	PNC assessment	Health worker	The system to suggest appropriate treatments/ investigations based on data entered	I can provide appropriate services according to the latest clinical protocols
PNC.FXNREQ.037	PNC assessment	Health worker	The system to highlight abnormal values	I can identify critical health issues or errors (if a data-entry issue)
PNC.FXNREQ.038	Counselling, management and treatment	Health worker	The system to provide context-sensitive, real-time decision support in response to entry of clinical data (alerts, advice, resources)	I can provide appropriate services according to the latest clinical protocols
PNC.FXNREQ.039	Counselling, management and treatment	Health worker	To provide decision support as appropriate for users based on data entered	I can provide appropriate services according to the latest clinical protocols
PNC.FXNREQ.040	Counselling, management and treatment	Health worker	To provide context-specific information in response to the entry of clinical data	I can provide appropriate services according to the latest clinical protocols
PNC.FXNREQ.041	Non-emergency referral	Health worker	To provide a referral for non-emergency services (e.g. immunization, family planning, appointment with specialist)	I can provide appropriate services according to the latest clinical protocols
PNC.FXNREQ.042	Emergency referral	Health worker	<i>See functional requirements for referral (Business process C)</i>	N/A
PNC.FXNREQ.043	Scheduling	Health worker	To see a schedule of available days	I can notify the woman/carers when to return for subsequent contact
PNC.FXNREQ.044	Scheduling	Health worker	To be able to input custom schedules to allow for contacts on specific days and times, account for holidays, etc.	I can inform the woman/carers of the next contact

Requirement ID	Activity ID and description	As a ...	I want ...	So that ...
PNC.FXNREQ.046	Scheduling	Health worker	To indicate (based on protocol) the preferred days for follow-up contact	The woman and/or newborn/infant receives services according to recommended contact schedule
PNC.FXNREQ.047	Scheduling	Health worker	To record identification and tracking information (e.g. mobile phone number) for client in schedule	I can follow up and remind the woman/carer when to return for services
PNC.FXNREQ.048	Scheduling	Health worker or clerk	A list or roster of all postpartum women and newborns/infants	I know which women and/or newborns/infants to follow up or are due for services
PNC.FXNREQ.049	Scheduling	Health worker or clerk	Tools to search, sort and filter the postnatal database	I can manage and follow up women and newborns/infants requiring services
PNC.FXNREQ.051	Woman and/or newborn/infant returns	Health worker	Ability to “check in” a woman and/or newborn/infant for a scheduled contact	I can initiate services at the follow-up contact
PNC.FXNREQ.052	Woman and/or newborn/infant returns	Health worker	To identify the date of the last attended contact	I can ensure the woman and/or newborn/infant is receiving services according to recommended contact schedule
Business process C: Emergency referral				
PNC.FXNREQ.053	In-facility referral	Health worker	To be able to find out if the required emergency service may be available within the same health facility	I can refer my client(s) to another department or ward in this same health facility to receive the appropriate services
PNC.FXNREQ.054	Discuss referral location options	Health worker	Find facilities that can provide the service or method that mine cannot provide	The client(s) do(es) not travel to a facility that cannot help them
PNC.FXNREQ.055	Contact referral health facility	Health worker	To have a list of the contact information for referral facilities	I can easily contact the facility when making the referral arrangements and ensure the facility can receive the patient
PNC.FXNREQ.056	Organize transport and provide information to referral health facility	Health worker	To share the client's/clients' health records with the referral facility	They can provide the care my client(s) need(s)
PNC.FXNREQ.057	Receive client information	Health worker	To know what care the client(s) received at the referral facility	I can provide appropriate care if my client(s) come(s) back to my facility
Business process D: PNC follow-up contact				
PNC.FXNREQ.058	Registration	Health worker or clerk	See functional requirements above for registration (Business process A)	N/A
PNC.FXNREQ.059	Review information from past PNC contact(s)	Health worker or clerk	See functional requirements above for PNC contact within 24 hours after birth (Business process B)	N/A
PNC.FXNREQ.060	PNC assessment	Health worker	See functional requirements above for PNC contact within 24 hours after birth (Business process B)	N/A

Requirement ID	Activity ID and description	As a ...	I want ...	So that ...
PNC.FXNREQ.061	Counselling, management and treatment	Health worker	See functional requirements above for PNC contact within 24 hours after birth (Business process B)	N/A
PNC.FXNREQ.062	Non-emergency referral	Health worker	See functional requirements above for PNC contact within 24 hours after birth (Business process B)	N/A
PNC.FXNREQ.063	Emergency referral	Health worker	See functional requirements for referral (Business process C)	N/A
Business process E: Reporting on aggregate indicators				
PNC.FXNREQ.066	Aggregate reporting and data use	Facility staff member	To produce a range of standard and ad hoc reports and analyses	I am able to use data collected at the facility, including for service delivery (beyond reporting purposes alone)
PNC.FXNREQ.067	Aggregate reporting and data use	Health worker (e.g. nurse/midwife)	To view data on my own performance and service delivery	I can understand trends and challenges and track my own performance over time

^a The term “client” is used for the registration workflow as this is applicable to other health domains, and “woman and/or newborn/infant” is used for the PNC-specific business processes.

8.2 Non-functional requirements

Table 18. Non-functional requirements

Requirement ID	Category	Non-functional requirement
PNC.NFXNREQ.001	Security – confidentiality	Provide password-protected access for authorized users
PNC.NFXNREQ.002	Security – confidentiality	Provide a means to ensure confidentiality and privacy of personal health information
PNC.NFXNREQ.003	Security – confidentiality	Provide ability for allowed users to view confidential data
PNC.NFXNREQ.004	Security – confidentiality	Anonymize data that is exported from the system
PNC.NFXNREQ.005	Security – confidentiality	Prevent remembering username and password
PNC.NFXNREQ.006	Security – confidentiality	Automatically log out the user after specified time of inactivity
PNC.NFXNREQ.007	Security – confidentiality	Provide encrypted communication between components
PNC.NFXNREQ.008	Security – authentication	Notify the user to change their password the first time they log in
PNC.NFXNREQ.009	Security – authentication	Adhere to complex password requirements
PNC.NFXNREQ.010	Security – authentication	Provide a mechanism to securely change a user’s password
PNC.NFXNREQ.011	Security – authentication	Notify the user of password change to their account
PNC.NFXNREQ.012	Security – authentication	Reset a user’s password in a secure manner

Requirement ID	Category	Non-functional requirement
PNC.NFXNREQ.013	Security – authentication	Lock a user out after a specified number of wrong password attempts
PNC.NFXNREQ.014	Security – authentication	Notify a user if their account is locked due to wrong password attempts
PNC.NFXNREQ.015	Security – authentication	Provide role-based access to the system
PNC.NFXNREQ.016	Security – audit trail and logs	Log system logins and logouts
PNC.NFXNREQ.017	Security – audit trail and logs	Record all authentication violations
PNC.NFXNREQ.018	Security – audit trail and logs	Log all activities performed by the user, including date-and-time stamp
PNC.NFXNREQ.019	Security – audit trail and logs	Log access to views of individual client records
PNC.NFXNREQ.020	Security – audit trail and logs	Log access to data summaries, reports, analysis and visualization features
PNC.NFXNREQ.021	Security – audit trail and logs	Log exchange of data with other systems
PNC.NFXNREQ.022	Security – audit trail and logs	Generate analysis of the usage of different system features and reports
PNC.NFXNREQ.023	Security – audit trail and logs	Log all data and system errors
PNC.NFXNREQ.024	Security – user management	Allow user with permission to create a new user and temporary password
PNC.NFXNREQ.025	Security – user management	Provide role-based access
PNC.NFXNREQ.026	Security – user management	Allow roles to be associated with specific geographical areas and/or health facilities
PNC.NFXNREQ.027	Security – user management	Allow cascading user management and assignment of roles
PNC.NFXNREQ.028	Security – user management	Allow user to change their own password
PNC.NFXNREQ.029	Security – user management	Allow admin user to request password reset
PNC.NFXNREQ.030	Security – user management	Notify the user to regularly change their password
PNC.NFXNREQ.031	Security – user management	Allow each user to be assigned to one or more roles
PNC.NFXNREQ.032	Security – user management	Support definitions of unlimited roles and assigned levels of access, viewing, entry, editing and auditing
PNC.NFXNREQ.033	System requirements – general	Provide a unique version number for each revision
PNC.NFXNREQ.034	System requirements – general	Enable earlier versions of a record to be recoverable
PNC.NFXNREQ.035	System requirements – general	Enable deployment in an environment subject to power loss
PNC.NFXNREQ.036	System requirements – general	Work in an environment that is subject to loss of connectivity
PNC.NFXNREQ.037	System requirements – general	Generate IDs that are unique across different installations or sites
PNC.NFXNREQ.038	System requirements – general	Report version number when saving data to the database
PNC.NFXNREQ.039	System requirements – general	Be designed to be flexible enough to accommodate necessary changes in the future
PNC.NFXNREQ.040	System requirements – general	Allow for offline and online functionality
PNC.NFXNREQ.041	System requirements – general	Show the number of records that are not yet synchronized

Requirement ID	Category	Non-functional requirement
PNC.NFXNREQ.042	System requirements – general	Have ability to easily back up information
PNC.NFXNREQ.043	System requirements – general	Warn user if no valid backup for more than a predefined number of days
PNC.NFXNREQ.044	System requirements – general	Must have the ability to store images and other unstructured data
PNC.NFXNREQ.045	System requirements – scalability	Scalable to accommodate new demands
PNC.NFXNREQ.046	System requirements – scalability	Be able to accommodate at least [x number of] health facilities
PNC.NFXNREQ.047	System requirements – scalability	Be able to accommodate at least [x number of] concurrent users
PNC.NFXNREQ.048	System requirements – usability	Be user-friendly for people with low computer literacy
PNC.NFXNREQ.049	System requirements – usability	Provide informative error messages and tool-tips
PNC.NFXNREQ.050	System requirements – usability	Alert the user when navigating away from a form without saving
PNC.NFXNREQ.051	System requirements – usability	Support real-time data-entry validation and feedback to prevent data-entry errors from being recorded
PNC.NFXNREQ.052	System requirements – usability	Simplify data recording through predefined drop-down menu or searchable lists, radio buttons, check boxes
PNC.NFXNREQ.053	System requirements – usability	Support multiple languages
PNC.NFXNREQ.054	System requirements – usability	Use industry standard user interface practices and apply them consistently throughout the system
PNC.NFXNREQ.055	System requirements – usability	Easy to learn and intuitive to enable user to navigate between pages
PNC.NFXNREQ.056	System requirements – usability	Provide guidance to users to better support clinical guidelines and best clinical practices
PNC.NFXNREQ.057	System requirements – usability	Be reliable and robust (minimize the number of system crashes)
PNC.NFXNREQ.058	System requirements – usability	Adjust display to fit small screens (e.g. mobile phones)
PNC.NFXNREQ.059	System requirements – configuration	Configure the system centrally
PNC.NFXNREQ.060	System requirements – configuration	Configure business rules in line with guidelines and standard operating procedures (SOPs)
PNC.NFXNREQ.061	System requirements – configuration	Configure error messages
PNC.NFXNREQ.062	System requirements – configuration	Configure workflows and business rules to accommodate differences between facilities
PNC.NFXNREQ.063	System requirements – interoperability	Communicate with external systems through mediators
PNC.NFXNREQ.064	System requirements – interoperability	Provide access to data through application programming interfaces (APIs)
PNC.NFXNREQ.065	System requirements – interoperability	Be interoperable with external systems through mediators
PNC.NFXNREQ.066	System requirements – interoperability	Link with insurance systems to verify eligibility and submit claims
PNC.NFXNREQ.067	System requirements – interoperability	Exchange data with other approved systems
PNC.NFXNREQ.068	System requirements – interoperability	Accept data from multiple input methods including paper, geocoding (GPS)
PNC.NFXNREQ.069	System requirements – hardware and connectivity	Allow for data exchange and efficient synchronization across multiple facilities and points of service when Internet is available, even when it is intermittent and slow

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Annex. Guidance for adding data elements to or amending existing data elements in the data dictionary

When adapting the data dictionary, data elements may need to be modified or added as a result of the structure of existing paper registers or local reporting requirements. If starting from paper-based registers and forms, you can find additional guidance in the *Digital transformation handbook for primary health care (1)*. The following is an overview of the data mapping to provide a template for standardizing the data dictionary. When amending the wording of data elements, it is important to ensure that the standard terminology codes still reflect the data element as originally intended.

What to note	Description
Activity ID	The ID number of the task in the workflow in which the data element is collected. This will denote the point in time at which this data element is collected.
Form ID	If the existing paper register or forms have an existing identification number, that should be noted as the Form ID. Indicate the Form ID in which the data element appears. This is important for ensuring that the design of the digital system has taken into account all the required paper forms and data elements on those paper forms.
Form data element label	List the label of the data element as written on the original form (or translated as closely as possible). This will be key in keeping track of which data elements from the original paper forms are duplicated. Note that duplicate data fields could have been included purposely in multiple forms (e.g. name, date of birth, village) as a means to identify an individual client.
Data element label	The label of the data element written in a way in which the end-users can easily understand, e.g. “education level”, “weight”, “height”, “reason(s) for coming to facility”. The data element label in this column is what will be used in the digital form as the digital register should not simply replace the paper registers but should also streamline processes and link duplicated data elements.
Description and definition	The description and definition of the data element, including any units that define the field (e.g. weight in kilograms [kg]). Provide a clear explanation of what this data field is requesting. This definition will be key for streamlining (e.g. resolving duplicate data elements) and detailing how required calculations are documented. Data element labels may differ across paper forms, so it is important to have a clear definition of this specific data element so that data elements with the same definition can be reconciled for one-time data entry. Conversely, some data elements with the same data element labels may in fact mean different things. This would require a change in the data element labels so that data entry is done accurately.
Multiple choice	If the data element is indicative of a multiple-choice question (e.g. symptoms), then indicate the type of multiple-choice question here. The types would be: » <i>Select one</i> (i.e. only one input can be chosen) » <i>Select all that apply</i> (i.e. more than one input option can be chosen) Each individual answer option should be listed in the Input options column and be classified with one of the Data types listed below.

What to note	Description
Data type	The data types are (2): » <i>Boolean</i> (i.e. true/false, yes/no) » <i>String</i> (i.e. a sequence of Unicode characters – e.g. name) » <i>Date</i> (e.g. date of birth) – used when only the date is recorded » <i>Time</i> (e.g. time of delivery) – used when only the time is recorded » <i>DateTime</i> (e.g. appointment) – used when the date and time are recorded » <i>ID</i> (e.g. unique identifier assigned to the health-service user) » <i>Quantity</i> – a number that is associated with a unit of measure outlined in the standard for Unified Code for Units of Measure (UCUM); quantities include any number that is associated with a unit, such as “number of past pregnancies”, where “past pregnancies” is the unit of measure (3) (if the data type is a <i>Quantity</i> there should be an associated sub-type listed in the Quantity sub-type column) » <i>Signature</i> (e.g. supervisor’s approval) – an electronic representation of a signature that is either cryptographic or a graphical image that represents a signature or a signature process » <i>Attachment</i> (e.g. image) – additional data content defined in other formats » <i>Coding</i> (e.g. symptoms, reason for coming to the facility, danger signs) – multiple-choice data elements for which the input options are <i>Codes</i> (e.g. pregnant, HIV positive, combined pill) – data elements that are input options to multiple-choice data elements, which are none of the above data types.
Input options	Input options are used for multiple-choice fields only. Write the list of responses from which the health worker may select. Each of these options should be labelled with a Data type as indicated above. For other fields, leave this column blank.
Calculation	If a calculation is needed to define the data element, write the formula here. Leave this column blank if no calculation is needed. Write the formula using standard mathematical symbols and the data element label included in the formula (e.g. for the body mass index [BMI] calculation, “weight/[height ²]”).
Required	Note whether this field is: » Required – R » Optional – O » Conditional on answers from other data fields – C
Reason for requiring data	If this field is required (R), state the reason here – whether for: » accountability for national-level reporting » service delivery or clinical decision-making » client identification. The digital system should not simply replace your paper registers, but it should also streamline processes; thus, it is important to understand why a certain data field is <i>actually</i> required and seek opportunities to optimize data flows. Given the high volume of data collection required of health-service providers, it might be better to remove a data entry field if it serves no real purpose for the clinician, public health reporting or any other identified purpose.

What to note	Description
Explain conditionality	If this field is conditional on answers from other data fields (C), denote what the conditionality is here. Conditionality helps to define the rules that govern the presence or absence of a data element based on certain criteria. This is common for data elements that are a part of follow-up questions. For example, if the input of one data element field is true, then some additional data inputs may be required.
Linked to aggregate indicator	List the indicators here if this data element contributes to an aggregate indicator. If the data element does not contribute to calculation of an aggregate indicator, leave this column blank. If the data element does not contribute to an aggregate indicator and is not needed for service delivery, consider removing it as a data field when designing the digital system. This would reduce the burden of data collection for health workers.
Notes	If there is an issue or inconsistency in how a data field is defined, make a note of the issue here. Irregularities and inconsistencies will need to be resolved at a later stage through a process of team discussion and triangulation. This column should also be used for any other notes, annotations or communication messages within the team.
Mapping(s) to standardized classifications and terminologies (3)	A column should be added for each classification or terminology code system (e.g. ICD-11, SNOMED, LOINC) the digital system is planned to use and interoperate with. The code used for each data element should be logged in these columns. This is a highly resource-intensive, but necessary, task. Any existing standardized code systems that can be used, should be used for the purposes of interoperability so that data can be represented in a consistent manner and exchanged with any other critical health information systems (e.g. lab systems, supply chain systems, reporting systems). This part can also be done through a terminology service.

ICD: International Classification of Diseases; ID: identification; LOINC: Logical Observation Identifiers Names and Codes; SNOMED: Systematized Nomenclature of Medicine.

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